

A BELLMAN-FUNCTION BOUND FOR THE BEURLING–AHLFORS TRANSFORM

HYRA

ABSTRACT. We prove the uniform estimate

$$\|B\|_{L^p(\mathbb{C}) \rightarrow L^p(\mathbb{C})} \leq 1.523958 (p^* - 1), \quad 1 < p < \infty,$$

for the Beurling–Ahlfors transform. The argument combines a heat-extension representation with an explicit Bellman-function criterion adapted to the conformal martingale structure of the transform. The principal new ingredients are a two-sided kinematic constraint that reduces the drift calculation to one negative-semidefinite 2×2 matrix, and two explicit generalized polynomial Bellman profiles at $p = 16/3$ and $p = 15/2$. Their drift and majorization properties are certified by exact polynomial identities and rational inequalities. Interpolation with the L^2 isometry and a self-contained Burkholder-type bound yields the stated constant.

INTRODUCTION

The Beurling–Ahlfors transform is the singular integral

$$Bf(z) = -\frac{1}{\pi} \text{p. v.} \int_{\mathbb{C}} \frac{f(w)}{(z-w)^2} dm(w).$$

For $1 < p < \infty$, write $p^* = \max\{p, p/(p-1)\}$. Iwaniec’s conjecture asserts that $\|B\|_p = p^* - 1$; see [4]. Bañuelos and Janakiraman proved the uniform upper bound $1.575(p^* - 1)$ by exploiting conformal martingales [1]. The present paper lowers that coefficient.

Main Theorem. For every $1 < p < \infty$,

$$\|B\|_{L^p(\mathbb{C}) \rightarrow L^p(\mathbb{C})} \leq 1.523958 (p^* - 1).$$

Consequently, if

$$C_{\text{BA}} := \sup_{1 < p < \infty} \frac{\|B\|_{L^p \rightarrow L^p}}{p^* - 1},$$

then $C_{\text{BA}} \leq 1.523958$.

The proof follows the heat-extension and space-time Brownian-motion framework of [2], but no quantitative martingale inequality is imported. Instead, the necessary martingale estimate is proved from an explicit radial Bellman function. The structural step is a two-sided inequality $|T - P| \leq W$, which permits the angular variable to be eliminated without a case split. This leads to a pointwise drift criterion expressed by a single 2×2 matrix. Generalized polynomial profiles with real exponents then produce the endpoint estimates

$$\|B\|_{16/3} \leq 22830^{3/16}, \quad \|B\|_{15/2} \leq 22865000^{2/15}.$$

Together with $\|B\|_2 = 1$, a self-contained estimate $\|B\|_p \leq \sqrt{2p(p-1)}$ for $p \geq 2$, duality, and Riesz–Thorin interpolation, these endpoints imply the main theorem.

All profile identities and all numerical comparisons used in the proof are exact. The only external inputs are standard results from Fourier analysis, functional analysis, and stochastic calculus, together with the classical multiplier identification of B . The use of compactly supported mollifiers below makes the regularization step entirely local and avoids any hidden tail estimate.

Date: August 18, 2026.

2020 Mathematics Subject Classification. Primary 42B20; Secondary 60G44.

Key words and phrases. Beurling–Ahlfors transform, Bellman function, conformal martingale, heat extension, singular integral, Riesz–Thorin interpolation.

0. NOTATION AND CONVENTIONS

- m = Lebesgue measure on $\mathbb{C} \simeq \mathbb{R}^2$; we write dx for it and identify $z = x_1 + ix_2 \leftrightarrow x = (x_1, x_2)$. All functions are complex valued unless stated otherwise. $\|f\|_q = (\int_{\mathbb{R}^2} |f|^q dx)^{1/q}$.
- Fourier transform: $\hat{f}(\xi) = \int_{\mathbb{R}^2} f(x) e^{-2\pi i x \cdot \xi} dx$, so that Plancherel reads $\int f \bar{g} dx = \int \hat{f} \bar{\hat{g}} d\xi$ and $\widehat{\partial_j f}(\xi) = 2\pi i \xi_j \hat{f}(\xi)$. We also identify $\xi = \xi_1 + i\xi_2 \in \mathbb{C}$ when convenient; $\bar{\xi} = \xi_1 - i\xi_2$, $|\xi|^2 = \xi_1^2 + \xi_2^2$.
- Heat semigroup: $G_t(x) = (4\pi t)^{-1} e^{-|x|^2/(4t)}$ for $t > 0$ (a probability density on $\mathbb{R}^2$), and for $f \in C_c^\infty(\mathbb{R}^2)$

$$U(x, t) := (e^{t\Delta} f)(x) = (f * G_t)(x), \quad U(x, 0) = f(x).$$

Then $\partial_t U = \Delta U$, $\hat{U}(\xi, t) = e^{-4\pi^2 |\xi|^2 t} \hat{f}(\xi)$, $U \in C^\infty(\mathbb{R}^2 \times [0, \infty))$ (differentiation under the integral sign is legitimate because $f \in C_c^\infty$ and $\partial^\alpha U = (\partial^\alpha f) * G_t$), and by Young's inequality $\|\partial^\alpha U(\cdot, t)\|_q \leq \|\partial^\alpha f\|_q$ for every multi-index α , every $q \in [1, \infty]$ and every $t \geq 0$.

- For $v, w \in \mathbb{C}^2$ we write $v \cdot w = \sum_j v_j w_j$ (**bilinear**, no conjugation) and $|v|^2 = \sum_j |v_j|^2$; thus $|v \cdot w| \leq |v| |w|$. For a real vector $h \in \mathbb{R}^d$, $|h|$ is the Euclidean norm and $h \cdot k$ the real inner product.
- $\mathbb{C}$ is identified with $\mathbb{R}^2$ as a real Euclidean space whenever a complex quantity is fed into a real function; the modulus of a complex number equals the Euclidean norm of the corresponding real vector, and for $\zeta, \eta \in \mathbb{C}$ the real inner product is $\zeta \cdot \eta = \operatorname{Re}(\zeta \bar{\eta})$.
- $\operatorname{artanh} x = \frac{1}{2} \ln \frac{1+x}{1-x}$ for $|x| < 1$.

The multiplier form of B . We use the classical identity

$$\widehat{Bf}(\xi) = \frac{\bar{\xi}}{\xi} \hat{f}(\xi) = \frac{\bar{\xi}^2}{|\xi|^2} \hat{f}(\xi), \quad \xi \neq 0, \quad (0.1)$$

initially for $f \in C_c^\infty(\mathbb{R}^2)$. In particular, Plancherel and $|\bar{\xi}/\xi| = 1$ give

$$\|B\|_{L^2 \rightarrow L^2} = 1. \quad (0.2)$$

All other L^p -extensions used below are obtained from the a priori estimates proved in this paper. More precisely, an estimate on C_c^∞ gives a unique bounded extension by density. If bounds are available at two exponents, the corresponding extensions agree on their intersection because C_c^∞ is dense there for the sum of the two norms. Thus the compatible form of the Riesz–Thorin theorem applies without assuming qualitative L^q -boundedness in advance.

1. REDUCTIONS AND THE HEAT REPRESENTATION

1.1. Reduction to $p \geq 2$.

Lemma 1.1. *For $f, g \in C_c^\infty(\mathbb{R}^2)$ one has $\int (Bf)g dx = \int f (Bg) dx$.*

Proof. For $F, G \in L^2$, Plancherel in the bilinear form reads $\int FG dx = \int \hat{F}(\xi) \hat{G}(-\xi) d\xi$ (apply $\int F \bar{H} = \int \hat{F} \bar{\hat{H}}$ with $H = \bar{G}$, whose transform is $\hat{H}(\xi) = \overline{\hat{G}(-\xi)}$). Hence, with $m(\xi) = \bar{\xi}/\xi$,

$$\int (Bf)g dx = \int m(\xi) \hat{f}(\xi) \hat{g}(-\xi) d\xi, \quad \int f (Bg) dx = \int \hat{f}(\xi) m(-\xi) \hat{g}(-\xi) d\xi.$$

Since $m(-\xi) = \overline{(-\xi)}/(-\xi) = \bar{\xi}/\xi = m(\xi)$, the two agree. $\square$

Lemma 1.2 (duality). *Let $1 < q < \infty$, $q' = q/(q-1)$, let $h \in L_{\text{loc}}^1(\mathbb{R}^2)$ and $M < \infty$ be such that $|\int h \bar{g} dx| \leq M \|g\|_q$ for every $g \in C_c^\infty$. Then $h \in L^{q'}$ and $\|h\|_{q'} \leq M$.*

Proof. The map $g \mapsto \int h \bar{g} dx$ is a conjugate-linear functional on the dense subspace C_c^∞ of L^q , of norm $\leq M$; extend it continuously to L^q . Since $1 < q < \infty$, the dual of L^q is $L^{q'}$, so there is $H \in L^{q'}$ with $\|H\|_{q'} \leq M$ and $\int h \bar{g} = \int H \bar{g}$ for every $g \in C_c^\infty$. Then $h - H \in L_{\text{loc}}^1$ annihilates C_c^∞ , hence $h = H$ a.e., and $\|h\|_{q'} = \|H\|_{q'} \leq M$. $\square$

Lemma 1.3 (transposition). *Suppose $\|Bg\|_q \leq M_q \|g\|_q$ for all $g \in C_c^\infty$ and some $1 < q < \infty$. Then $\|Bf\|_{q'} \leq M_q \|f\|_{q'}$ for all $f \in C_c^\infty$, where $1/q + 1/q' = 1$. Consequently $\|B\|_{L^p \rightarrow L^p} = \|B\|_{L^{p'} \rightarrow L^{p'}}$ for all $1 < p < \infty$.*

Proof. Let $f \in C_c^\infty$; then $Bf \in L^2 \subset L_{\text{loc}}^1$. For every $g \in C_c^\infty$ we also have $\bar{g} \in C_c^\infty$, so Lemma 1.1 (applied to the pair $f, \bar{g}$) and Hölder give

$$\left| \int (Bf) \bar{g} dx \right| = \left| \int f (B\bar{g}) dx \right| \leq \|f\|_{q'} \|B\bar{g}\|_q \leq M_q \|f\|_{q'} \|g\|_q.$$

Lemma 1.2 (with $h = Bf$, $M = M_q \|f\|_{q'}$) gives $\|Bf\|_{q'} \leq M_q \|f\|_{q'}$. By density $\|B\|_{q'} \leq \|B\|_q$, and by symmetry (replace q by q') the reverse inequality holds as well. $\square$

Since $p^* - 1 = q - 1$ both for $p = q \geq 2$ and for $p = q' \leq 2$, Lemma 1.3 shows:

It suffices to prove $\|Bf\|_p \leq \Lambda(p) \|f\|_p$ **for all** $f \in C_c^\infty$ **and all** $p \geq 2$, where $\Lambda(p)$ is the desired bound; the case $1 < p < 2$ then follows with $\Lambda(p')$.

For $p \geq 2$ we write $p' = p/(p-1) \in (1, 2]$ and $q := p-1 \geq 1$.

1.2. The matrix A and conformality. Let

$$w := (1, -i) \in \mathbb{C}^2, \quad A := w w^T = \begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix} \in \mathbb{C}^{2 \times 2}.$$

For a real vector $\xi = (\xi_1, \xi_2)$,

$$\langle A\xi, \xi \rangle := \sum_{j,k} A_{jk} \xi_j \xi_k = (w \cdot \xi)^2 = (\xi_1 - i\xi_2)^2 = \bar{\xi}^2,$$

so by (0.1) the symbol of B is exactly $\langle A\xi, \xi \rangle / |\xi|^2$.

Lemma 1.4 (structure of A). *Let $v = (v_1, v_2) \in \mathbb{C}^2$ and put $c := w \cdot v = v_1 - iv_2 \in \mathbb{C}$. Then*

$$Av = (c, -ic), \quad |Av|^2 = 2|c|^2 \leq 4|v|^2. \quad (1.1)$$

Moreover, identifying $\mathbb{C} \simeq \mathbb{R}^2$, the two components $k_1 := c$ and $k_2 := -ic$ of Av satisfy, for every unit vector $\hat{e} \in \mathbb{R}^2$,

$$\sum_{j=1}^2 (k_j \cdot \hat{e})^2 = |c|^2 = \frac{1}{2} \sum_{j=1}^2 |k_j|^2, \quad \sum_{j=1}^2 |k_j - (k_j \cdot \hat{e})\hat{e}|^2 = |c|^2. \quad (1.2)$$

Proof. $Av = w(w \cdot v) = cw = (c, -ic)$, and $|Av|^2 = |c|^2 + |-ic|^2 = 2|c|^2$; also $|c| = |v_1 - iv_2| \leq |v_1| + |v_2| \leq \sqrt{2}|v|$, whence $|Av|^2 \leq 4|v|^2$. For (1.2): write $\hat{e} \in \mathbb{R}^2$ as a unit complex number e , so that $k \cdot \hat{e} = \text{Re}(ke)$ for $k \in \mathbb{C}$. Then

$$(k_1 \cdot \hat{e})^2 + (k_2 \cdot \hat{e})^2 = (\text{Re}(ce))^2 + (\text{Re}(-ice))^2 = (\text{Re}(ce))^2 + (\text{Im}(ce))^2 = |ce|^2 = |c|^2,$$

using $\text{Re}(-i\zeta) = \text{Im}(\zeta)$ and $|e| = 1$. Since $|k_j|^2 = |c|^2$ for $j = 1, 2$, we get $\sum_j |k_j|^2 = 2|c|^2$ and $\sum_j |k_j - (k_j \cdot \hat{e})\hat{e}|^2 = \sum_j (|k_j|^2 - (k_j \cdot \hat{e})^2) = 2|c|^2 - |c|^2 = |c|^2$. $\square$

The identity (1.2) expresses the fact that $\{k_1, k_2\}$ is an *orthogonal pair of equal length*: $k_2 = -ik_1$ is k_1 rotated by $-\pi/2$. Any such pair distributes its total energy $2|c|^2$ **isotropically**: exactly half of it lies along any prescribed direction and half orthogonally to it. The inequality $|Av| \leq 2|v|$ in (1.1) will be used only qualitatively (to know that certain L^2 -quantities are finite); the *quantitative* input is (1.2).

1.3. Heat representation of the bilinear form.

Lemma 1.5. *Let $f, g \in C_c^\infty(\mathbb{R}^2)$, $U = e^{t\Delta}f$, $V = e^{t\Delta}g$. Then*

$$\int_{\mathbb{R}^2} (Bf) \bar{g} dx = 2 \int_0^\infty \int_{\mathbb{R}^2} (A \nabla U(x, t)) \cdot \overline{\nabla V(x, t)} dx dt, \quad (1.3)$$

where $\nabla U = (\partial_1 U, \partial_2 U) \in \mathbb{C}^2$ and the dot is the bilinear pairing of Section 0 (so the right side is $\sum_{j,k} A_{jk} \iint \partial_k U \bar{\partial}_j V$). Moreover $\int_0^\infty \int |\nabla U| |\nabla V| dx dt < \infty$ and

$$\int_0^\infty \int_{\mathbb{R}^2} |\nabla U|^2 dx dt = \frac{1}{2} \|f\|_2^2. \quad (1.4)$$

Proof. Fix $t > 0$. By Plancherel, for each j, k ,

$$\int \partial_k U \overline{\partial_j V} dx = \int (2\pi i \xi_k \hat{U}) \overline{(2\pi i \xi_j \hat{V})} d\xi = 4\pi^2 \int \xi_j \xi_k e^{-8\pi^2 |\xi|^2 t} \hat{f} \hat{g} d\xi.$$

Summing against A_{jk} ,

$$\int (A \nabla U) \cdot \overline{\nabla V} dx = 4\pi^2 \int \langle A\xi, \xi \rangle e^{-8\pi^2 |\xi|^2 t} \hat{f} \hat{g} d\xi.$$

The double integral converges absolutely: $|\langle A\xi, \xi \rangle| = |\bar{\xi}^2| = |\xi|^2$ and $\int_0^\infty \int 4\pi^2 |\xi|^2 e^{-8\pi^2 |\xi|^2 t} |\hat{f}| |\hat{g}| d\xi dt = \frac{1}{2} \int |\hat{f}| |\hat{g}| d\xi < \infty$ (Tonelli; $\hat{f}, \hat{g}$ are Schwartz). Hence by Fubini and $\int_0^\infty e^{-8\pi^2 |\xi|^2 t} dt = (8\pi^2 |\xi|^2)^{-1}$,

$$2 \int_0^\infty \int (A \nabla U) \cdot \overline{\nabla V} dx dt = 2 \cdot \frac{4\pi^2}{8\pi^2} \int \frac{\langle A\xi, \xi \rangle}{|\xi|^2} \hat{f} \hat{g} d\xi = \int \frac{\bar{\xi}^2}{|\xi|^2} \hat{f} \hat{g} d\xi = \int (Bf) \bar{g} dx,$$

the last step by (0.1) and Plancherel. For the final assertions, Plancherel gives

$$\int_0^\infty \int |\nabla U|^2 dx dt = \int_0^\infty \int 4\pi^2 |\xi|^2 e^{-8\pi^2 |\xi|^2 t} |\hat{f}|^2 d\xi dt = \frac{1}{2} \|f\|_2^2 < \infty,$$

and similarly for V ; then Cauchy–Schwarz yields $\int_0^\infty \int |\nabla U| |\nabla V| \leq \frac{1}{2} \|f\|_2 \|g\|_2 < \infty$. $\square$

Thus everything is reduced to estimates of the form

$$\left| 2 \int_0^\infty \int (A \nabla U) \cdot \overline{\nabla V} dx dt \right| \leq \Lambda \|f\|_p \|g\|_{p'}, \quad p \geq 2, \quad (1.5)$$

which are proved in Section 3–Section 4.

2. ADMISSIBLE BELLMAN FUNCTIONS

Throughout this section $p \geq 2$ is a fixed real number.

2.1. The two-sided kinematic constraint.

Lemma 2.1. *Let $a, b \in \mathbb{C}$, put $c := a - ib$ and $T := |c|$, and let $\hat{e} \in \mathbb{R}^2$ be a unit vector. Set $h_1 := a$, $h_2 := b$ (viewed in $\mathbb{R}^2$) and*

$$P := \left(\sum_{j=1}^2 (h_j \cdot \hat{e})^2 \right)^{1/2}, \quad W := \left(\sum_{j=1}^2 |h_j - (h_j \cdot \hat{e}) \hat{e}|^2 \right)^{1/2}.$$

Then

$$|T - P| \leq W, \quad \text{equivalently} \quad T \leq P + W \quad \text{and} \quad P \leq T + W. \quad (2.1)$$

Proof. Let $\varepsilon \in \mathbb{C}$, $|\varepsilon| = 1$, correspond to $\hat{e}$, and set $a' := a\bar{\varepsilon} = a_1 + ia_2$, $b' := b\bar{\varepsilon} = b_1 + ib_2$ with $a_i, b_i \in \mathbb{R}$. Since $h \cdot \hat{e} = \operatorname{Re}(h\bar{\varepsilon})$ and $|h - (h \cdot \hat{e})\hat{e}| = |\operatorname{Im}(h\bar{\varepsilon})|$ for $h \in \mathbb{C}$,

$$P^2 = a_1^2 + b_1^2, \quad W^2 = a_2^2 + b_2^2.$$

Moreover $c\bar{\varepsilon} = a' - ib' = (a_1 + b_2) + i(a_2 - b_1)$, so, as vectors in $\mathbb{R}^2$,

$$T = |c| = |c\bar{\varepsilon}| = |(a_1 + b_2, a_2 - b_1)| = |\mathbf{u} + \mathbf{v}|, \quad \mathbf{u} := (a_1, -b_1), \quad \mathbf{v} := (b_2, a_2),$$

and $|\mathbf{u}| = P$, $|\mathbf{v}| = W$. The triangle inequality gives $T = |\mathbf{u} + \mathbf{v}| \leq |\mathbf{u}| + |\mathbf{v}| = P + W$ and $P = |\mathbf{u}| = |(\mathbf{u} + \mathbf{v}) - \mathbf{v}| \leq T + W$. $\square$

Only the first half of (2.1) is classical. The second half, $P \leq T + W$, is what allows us below to replace W^2 by $(T - P)^2$ **without a case distinction**, and hence to reduce the whole drift condition to the negative semidefiniteness of a single 2×2 matrix.

2.2. The drift criterion. Let $\varphi : [0, \infty)^2 \rightarrow \mathbb{R}$ be continuous, and set

$$u(x, y) := \varphi(|x|, |y|), \quad x, y \in \mathbb{R}^2.$$

Suppose φ is C^2 on the open quadrant $\{r > 0, s > 0\}$. There, with $\hat{x} = x/|x|$, $\hat{y} = y/|y|$, $r = |x|$, $s = |y|$, the classical Hessian of u is

$$D_{xx}^2 u = \varphi_{rr} \hat{x} \otimes \hat{x} + \frac{\varphi_r}{r} (I - \hat{x} \otimes \hat{x}), \quad D_{xy}^2 u = \varphi_{rs} \hat{x} \otimes \hat{y}, \quad D_{yy}^2 u = \varphi_{ss} \hat{y} \otimes \hat{y} + \frac{\varphi_s}{s} (I - \hat{y} \otimes \hat{y})$$

(these follow from $\partial_{x_i} |x| = \hat{x}_i$ and $\partial_{x_i} \hat{x}_j = (\delta_{ij} - \hat{x}_i \hat{x}_j)/r$, so that $\partial_{x_i} \partial_{x_j} u = \varphi_{rr} \hat{x}_i \hat{x}_j + \varphi_r (\delta_{ij} - \hat{x}_i \hat{x}_j)/r$, and similarly in y ; the mixed block is $\partial_{x_i} \partial_{y_j} u = \varphi_{rs} \hat{x}_i \hat{y}_j$). Hence for $h, k \in \mathbb{R}^2$, writing $h_{\parallel} := h \cdot \hat{x}$, $h_{\perp} := h - h_{\parallel} \hat{x}$, $k_{\parallel} := k \cdot \hat{y}$, $k_{\perp} := k - k_{\parallel} \hat{y}$,

$$\langle D^2 u(x, y)(h, k), (h, k) \rangle = \varphi_{rr} h_{\parallel}^2 + 2\varphi_{rs} h_{\parallel} k_{\parallel} + \varphi_{ss} k_{\parallel}^2 + \frac{\varphi_r}{r} |h_{\perp}|^2 + \frac{\varphi_s}{s} |k_{\perp}|^2. \quad (2.2)$$

Definition 2.2 (the drift matrix). For $r, s > 0$ put

$$\mathcal{A} := \varphi_{rr} + \frac{\varphi_r}{r}, \quad \mathcal{B} := |\varphi_{rs}| - \frac{\varphi_r}{r}, \quad \mathcal{C} := \varphi_{ss} + \frac{\varphi_s}{s} + \frac{\varphi_r}{r},$$

all evaluated at (r, s) , and call $\mathcal{Q} := \begin{pmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{B} & \mathcal{C} \end{pmatrix}$ the *drift matrix* of φ at (r, s) .

Proposition 2.3 (drift inequality). Let $x, y \in \mathbb{R}^2$ with $x \neq 0 \neq y$, and let $a, b \in \mathbb{C}$ be arbitrary. Put $c := a - ib$ and

$$v_1 := (a, c), \quad v_2 := (b, -ic) \in \mathbb{R}^2 \times \mathbb{R}^2 = \mathbb{R}^4.$$

Assume that at $(r, s) = (|x|, |y|)$ one has

$$\varphi_r \leq 0 \quad (D0)$$

and that the drift matrix is negative semidefinite, i.e.

$$\mathcal{A} \leq 0, \quad \mathcal{C} \leq 0, \quad \mathcal{B}^2 \leq \mathcal{A}\mathcal{C}. \quad (D1)$$

Then

$$\sum_{j=1}^2 \langle D^2 u(x, y) v_j, v_j \rangle \leq 0. \quad (2.3)$$

Proof. Write $h_1 = a$, $h_2 = b$, $k_1 = c$, $k_2 = -ic$; the j -th vector is $v_j = (h_j, k_j)$. Apply (2.2) to each j and sum. Let P, W be as in Lemma 2.1 with $\hat{e} = \hat{x}$, so $\sum_j h_{j\parallel}^2 = P^2$ and $\sum_j |h_{j\perp}|^2 = W^2$; and let $T = |c|$. Lemma 1.4, applied with $\hat{e} = \hat{y}$, gives

$$\sum_j k_{j\parallel}^2 = T^2, \quad \sum_j |k_{j\perp}|^2 = T^2.$$

By Cauchy–Schwarz, $|\sum_j h_{j\parallel} k_{j\parallel}| \leq (\sum_j h_{j\parallel}^2)^{1/2} (\sum_j k_{j\parallel}^2)^{1/2} = PT$. Therefore

$$\sum_j \langle D^2 u v_j, v_j \rangle \leq \varphi_{rr} P^2 + 2|\varphi_{rs}| PT + \left(\varphi_{ss} + \frac{\varphi_s}{s} \right) T^2 + \frac{\varphi_r}{r} W^2.$$

By (D0) the coefficient φ_r/r of W^2 is ≤ 0 , and by Lemma 2.1 $W^2 \geq (T - P)^2$; hence $\frac{\varphi_r}{r} W^2 \leq \frac{\varphi_r}{r} (T - P)^2$ and

$$\sum_j \langle D^2 u v_j, v_j \rangle \leq \left(\varphi_{rr} + \frac{\varphi_r}{r} \right) P^2 + 2 \left(|\varphi_{rs}| - \frac{\varphi_r}{r} \right) PT + \left(\varphi_{ss} + \frac{\varphi_s}{s} + \frac{\varphi_r}{r} \right) T^2 = \mathcal{A}P^2 + 2\mathcal{B}PT + \mathcal{C}T^2.$$

A real binary quadratic form $\mathcal{A}P^2 + 2\mathcal{B}PT + \mathcal{C}T^2$ with $\mathcal{A} \leq 0$, $\mathcal{C} \leq 0$ and $\mathcal{B}^2 \leq \mathcal{A}\mathcal{C}$ is negative semidefinite on all of $\mathbb{R}^2$: indeed if $\mathcal{A} < 0$ then $\mathcal{A}P^2 + 2\mathcal{B}PT + \mathcal{C}T^2 = \mathcal{A} \left(P + \frac{\mathcal{B}}{\mathcal{A}} T \right)^2 + \frac{\mathcal{A}\mathcal{C} - \mathcal{B}^2}{\mathcal{A}} T^2 \leq 0$, while if $\mathcal{A} = 0$ then $\mathcal{B}^2 \leq 0$ forces $\mathcal{B} = 0$ and the form is $\mathcal{C}T^2 \leq 0$. This proves (2.3). $\square$

2.3. Admissible functions.

Definition 2.4. Let $p \geq 2$ and $\Lambda > 0$. A function $\varphi : [0, \infty)^2 \rightarrow \mathbb{R}$ is called (p, Λ) -**admissible** if:

- **(A1)** φ is continuous on $[0, \infty)^2$, positively homogeneous of degree p (i.e. $\varphi(\lambda r, \lambda s) = \lambda^p \varphi(r, s)$ for $\lambda > 0$), and C^∞ on $\{r > 0, s > 0\}$;
- **(A2)** (*growth*) there are $M < \infty$ and $\theta \in [0, 1)$ with, for all $r, s > 0$ and $\sigma := r + s$,

$$|\varphi| \leq M\sigma^p, \quad |\varphi_r| + |\varphi_s| \leq M\sigma^{p-1} \left(\frac{\sigma}{s}\right)^\theta,$$

$$|\varphi_{rr}| + |\varphi_{rs}| + |\varphi_{ss}| + \left|\frac{\varphi_r}{r}\right| + \left|\frac{\varphi_s}{s}\right| \leq M\sigma^{p-2} \left(\frac{\sigma}{s}\right)^{1+\theta};$$

- **(A3)** (*drift*) (D0) and (D1) of Proposition 2.3 hold at every (r, s) with $r, s > 0$;
- **(A4)** (*majorization*) $\varphi(r, s) \geq s^p - \Lambda^p r^p$ for all $r, s \geq 0$;
- **(A5)** (*boundary sign*) $\varphi(r, 0) \leq 0$ for all $r \geq 0$.

Note $\sigma/s = 1 + r/s \geq 1$, so (A2) with $\theta = 0$ is the strongest form and reads $|\varphi_r| + |\varphi_s| \leq M\sigma^{p-1}$ and (second order) $\leq M\sigma^{p-2}(1 + r/s)$; larger $\theta < 1$ makes (A2) weaker. The single purpose of θ is to allow profiles whose s -derivatives blow up like $s^{-\theta}$ at $s = 0$; everything in Section 2.4 and Section 3 goes through unchanged for any $\theta < 1$.

We will prove in Section 3:

Bellman principle. If φ is (p, Λ) -admissible for some $p \geq 2$, then

$$\|B\|_{L^p \rightarrow L^p} \leq \Lambda.$$

Three admissible functions are exhibited in Section 4.

2.4. Regularity and mollification. Let φ be (p, Λ) -admissible with growth parameters (M, θ) , and let $u(x, y) = \varphi(|x|, |y|)$ on $\mathbb{R}^4$. We write $z = (x, y)$ and $s = |y|$.

Lemma 2.5. u is continuous on $\mathbb{R}^4$, and for some $M' = M'(M, p)$,

$$|u(z)| \leq M'|z|^p, \quad |\nabla u(z)| \leq M'|z|^{p-1} \left(\frac{|z|}{|y|}\right)^\theta \quad (y \neq 0). \quad (2.4)$$

The pointwise gradient and Hessian of u exist off the null set $\mathcal{N} := \{x = 0\} \cup \{y = 0\}$, belong to $L^1_{\text{loc}}(\mathbb{R}^4)$, and coincide with the distributional gradient and Hessian of u .

Proof. Continuity and the first bound in (2.4) are immediate from (A1)–(A2) (note $\sigma = |x| + |y| \leq \sqrt{2}|z|$ and $|z| \leq \sigma$). Off $\mathcal{N}$, u is smooth and $|\nabla u| \leq |\varphi_r| + |\varphi_s| \leq M\sigma^{p-1}(\sigma/s)^\theta$, which is the second bound in (2.4).

By (A2), $|D^2 u(z)| \leq C_p M\sigma^{p-2}(\sigma/s)^{1+\theta}$ off $\mathcal{N}$ (each entry of $D^2 u$ is a combination of $\varphi_{rr}, \varphi_{rs}, \varphi_{ss}, \varphi_r/r, \varphi_s/s$ with coefficients bounded by 1, by the display preceding (2.2)). Hence on the ball $\{|z| < R\}$,

$$|\nabla u(z)| \leq C(R)|y|^{-\theta}, \quad |D^2 u(z)| \leq C(R)|y|^{-1-\theta}.$$

Both are locally integrable on $\mathbb{R}^4$: in polar coordinates in the y -variable, $\int_{|y| < R} |y|^{-\alpha} dy = 2\pi R^{2-\alpha}/(2-\alpha) < \infty$ whenever $\alpha < 2$, and here $\alpha = \theta < 1$ resp. $\alpha = 1 + \theta < 2$.

Let $\zeta \in C_c^\infty(\mathbb{R}^4)$, supported in $\{|z| < R\}$, and let $\mathbf{a}, \mathbf{b}$ be two of the four coordinates. Let $\Omega_\delta := \{(x, y) : |x| > \delta, |y| > \delta\}$. On Ω_δ the function u is C^∞ , so two integrations by parts give

$$\int_{\Omega_\delta} u \partial_{\mathbf{a}} \partial_{\mathbf{b}} \zeta = \int_{\Omega_\delta} (\partial_{\mathbf{a}} \partial_{\mathbf{b}} u) \zeta + R_\delta,$$

where R_δ is a sum of boundary integrals over $\partial\Omega_\delta \subset (\{|x| = \delta\} \times \mathbb{R}^2) \cup (\mathbb{R}^2 \times \{|y| = \delta\})$ of terms $u(\partial\zeta)n$ and $(\partial u)\zeta n$. We estimate the two pieces.

On $\{|x| = \delta\} \cap \text{supp } \zeta$: the surface measure is $d\mathcal{H}^3 = \delta d\theta dy$ with $|y| < R$, of total mass $O(\delta)$. There $|u| \leq C(R)$ and $\int_{|y| < R} |\nabla u| dy \leq C(R) \int_{|y| < R} |y|^{-\theta} dy < \infty$, so both kinds of terms are $O(\delta)$.

On $\{|y| = \delta\} \cap \text{supp } \zeta$: the surface measure is $O(\delta)$; there $|u| \leq C(R)$ and $|\nabla u| \leq C(R)\delta^{-\theta}$, so the terms are $O(\delta)$ and $O(\delta^{1-\theta})$ respectively.

Since $\theta < 1$, $R_\delta \rightarrow 0$ as $\delta \downarrow 0$. Both integrands are dominated by L^1 functions ($|u| |\partial\zeta|$ is bounded with compact support; $|D^2 u| |\zeta|$ is in L^1 as shown), so the integrals converge to those over $\mathbb{R}^4$. Thus the distributional Hessian is the pointwise one. The same argument with one integration by parts (only the terms $u(\partial\zeta)n$ occur, of size $O(\delta)$) identifies the distributional gradient. $\square$

Lemma 2.6 (smoothed Bellman function). *Let $\rho \in C_c^\infty(\mathbb{R}^4)$ be nonnegative, supported in the unit ball, and normalized by $\int_{\mathbb{R}^4} \rho = 1$. Set $\rho_\varepsilon(z) = \varepsilon^{-4} \rho(z/\varepsilon)$ and $u_\varepsilon = u * \rho_\varepsilon$. Then:*

1. $u_\varepsilon \in C^\infty(\mathbb{R}^4)$; for some $C = C(M, p)$,

$$|u_\varepsilon(z)| \leq C(|z|^p + \varepsilon^p),$$

and ∇u_ε is bounded on compact subsets of $\mathbb{R}^4$.

2. For every $z \in \mathbb{R}^4$ and every $a, b \in \mathbb{C}$, with

$$v_1 = (a, a - ib), \quad v_2 = (b, -i(a - ib)),$$

one has

$$\sum_{j=1}^2 \langle D^2 u_\varepsilon(z) v_j, v_j \rangle \leq 0. \quad (2.5)$$

3. $u_\varepsilon \rightarrow u$ locally uniformly as $\varepsilon \downarrow 0$.

Proof. By Lemma 2.5, $u \in W_{\text{loc}}^{2,1}(\mathbb{R}^4)$ and has polynomial growth. Standard properties of convolution therefore give $u_\varepsilon \in C^\infty$,

$$D^2 u_\varepsilon = (D^2 u) * \rho_\varepsilon, \quad \nabla u_\varepsilon = (\nabla u) * \rho_\varepsilon.$$

Because ρ_ε is supported in $B(0, \varepsilon)$, the growth bound follows from $|u(z - \omega)| \leq C(|z| + |\omega|)^p$, and the local boundedness of the gradient is immediate from smoothness.

Fix z, a, b . For almost every ω , the point $z - \omega = (x', y')$ satisfies $x' \neq 0 \neq y'$, so Proposition 2.3 gives

$$\sum_{j=1}^2 \langle D^2 u(z - \omega) v_j, v_j \rangle \leq 0.$$

The vectors v_1, v_2 do not depend on the base point. Integrating the last inequality against the nonnegative kernel ρ_ε proves (2.5). Finally, continuity of u implies local uniform convergence under this compactly supported approximate identity. $\square$

2.5. The criterion in the variable $t = r/s$. It is convenient, for the explicit examples of Section 4, to record (A3) in terms of the profile of φ along $s = 1$. By homogeneity we may write

$$\varphi(r, s) = s^p f(t), \quad t := \frac{r}{s} \in (0, \infty), \quad f \in C^\infty(0, \infty). \quad (2.6)$$

Lemma 2.7. *With (2.6) and $r, s > 0$,*

$$\varphi_r = s^{p-1} f', \quad \varphi_{rr} = s^{p-2} f'', \quad \frac{\varphi_r}{r} = s^{p-2} \frac{f'}{t},$$

$$\varphi_{rs} = s^{p-2} [(p-1)f' - tf''], \quad \frac{\varphi_s}{s} = s^{p-2} [pf - tf'], \quad \varphi_{ss} = s^{p-2} [p(p-1)f - 2(p-1)tf' + t^2 f''],$$

all evaluated at $t = r/s$. Consequently

$$\mathcal{A} = \frac{s^{p-2}}{t} P_1, \quad \mathcal{C} = \frac{s^{p-2}}{t} P_2, \quad \text{and } \mathcal{B} = \frac{s^{p-2}}{t} P_3 \text{ provided } Q \geq 0,$$

where

$$P_1 := tf'' + f', \quad P_2 := p^2 tf - (2p-1)t^2 f' + t^3 f'' + f', \quad P_3 := t^2 f'' - (p-1)tf' - f',$$

$$Q := t^2 f'' - (p-1)tf'.$$

Hence, if for all $t > 0$

$$f' \leq 0, \quad Q \geq 0, \quad P_1 \leq 0, \quad P_2 \leq 0, \quad R := P_1 P_2 - P_3^2 \geq 0, \quad (2.7)$$

then (D0) and (D1) hold at every (r, s) with $r, s > 0$.

Proof. Differentiate (2.6), using $\partial_r t = 1/s$ and $\partial_s t = -t/s$: $\varphi_r = s^p f' \cdot \frac{1}{s} = s^{p-1} f'$ and $\varphi_{rr} = s^{p-1} f'' \cdot \frac{1}{s} = s^{p-2} f''$, giving also $\varphi_r/r = s^{p-1} f'/(st) = s^{p-2} f'/t$. Next

$$\varphi_s = ps^{p-1} f + s^p f' \cdot \left(-\frac{t}{s}\right) = s^{p-1} [pf - tf'],$$

so $\varphi_s/s = s^{p-2} [pf - tf']$. Differentiating $\varphi_r = s^{p-1} f'$ in s ,

$$\varphi_{rs} = (p-1)s^{p-2} f' + s^{p-1} f'' \cdot \left(-\frac{t}{s}\right) = s^{p-2} [(p-1)f' - tf''].$$

Differentiating $\varphi_s = s^{p-1} [pf - tf']$ in s and using $\partial_s(pf - tf') = pf' \cdot (-\frac{t}{s}) + \frac{t}{s} f' - tf'' \cdot (-\frac{t}{s}) = \frac{1}{s} [-(p-1)tf' + t^2 f'']$,

$$\varphi_{ss} = (p-1)s^{p-2} [pf - tf'] + s^{p-2} [-(p-1)tf' + t^2 f''] = s^{p-2} [p(p-1)f - 2(p-1)tf' + t^2 f''].$$

Now

$$\mathcal{A} = \varphi_{rr} + \frac{\varphi_r}{r} = s^{p-2} \left[f'' + \frac{f'}{t} \right] = \frac{s^{p-2}}{t} [tf'' + f'] = \frac{s^{p-2}}{t} \mathbf{P}_1,$$

$$\mathcal{C} = \varphi_{ss} + \frac{\varphi_s}{s} + \frac{\varphi_r}{r} = s^{p-2} \left[p(p-1)f - 2(p-1)tf' + t^2 f'' + pf - tf' + \frac{f'}{t} \right] = \frac{s^{p-2}}{t} [p^2 tf - (2p-1)t^2 f' + t^3 f'' + f'],$$

which is $\frac{s^{p-2}}{t} \mathbf{P}_2$. Finally $\mathbf{Q} \geq 0$ means $t^2 f'' - (p-1)tf' \geq 0$, i.e. (dividing by $t > 0$) $tf'' - (p-1)f' \geq 0$, i.e. $\varphi_{rs} = s^{p-2} [(p-1)f' - tf''] \leq 0$; in that case

$$\mathcal{B} = |\varphi_{rs}| - \frac{\varphi_r}{r} = s^{p-2} \left[tf'' - (p-1)f' - \frac{f'}{t} \right] = \frac{s^{p-2}}{t} [t^2 f'' - (p-1)tf' - f'] = \frac{s^{p-2}}{t} \mathbf{P}_3.$$

Since $s^{p-2}/t > 0$, the conditions $\mathcal{A} \leq 0$, $\mathcal{C} \leq 0$, $\mathcal{B}^2 \leq \mathcal{A}\mathcal{C}$ are equivalent to $\mathbf{P}_1 \leq 0$, $\mathbf{P}_2 \leq 0$, $\mathbf{P}_3^2 \leq \mathbf{P}_1\mathbf{P}_2$, and (D0) is $f' \leq 0$. $\square$

2.6. Generalized polynomial profiles.

Lemma 2.8. *Let $p \geq 2$, let $E \subset [2, p)$ be a finite set of **real** exponents, let $b_k \geq 0$ for $k \in E$, let $a_0 > 0$, and put*

$$f(t) = a_0 - \sum_{k \in E} b_k t^k \quad (t > 0), \quad \varphi(r, s) := s^p f(r/s) = a_0 s^p - \sum_{k \in E} b_k r^k s^{p-k}.$$

Then, with $\mathbf{P}_i, \mathbf{Q}$ as in Lemma 2.7,

$$\mathbf{P}_1 = - \sum_k k^2 b_k t^{k-1} \leq 0, \quad \mathbf{P}_3 = \sum_k k b_k t^{k-1} + \sum_k k(p-k) b_k t^k \geq 0,$$

$$\mathbf{P}_2 = p^2 a_0 t - \sum_k k b_k t^{k-1} - \sum_k (p-k)^2 b_k t^{k+1},$$

$$\mathbf{Q} = \sum_k k(p-k) b_k t^k \geq 0, \quad f' \leq 0.$$

Moreover φ extends continuously to $[0, \infty)^2$ with $\varphi(r, 0) = 0$ (so (A5) holds), is positively homogeneous of degree p , is C^∞ on $\{r > 0, s > 0\}$ — i.e. (A1) holds — and satisfies (A2) with

$$\theta := \max \left\{ 0, \max_{k \in E} (k+1-p) \right\} \in [0, 1).$$

Proof. $f' = - \sum k b_k t^{k-1} \leq 0$ and $f'' = - \sum k(k-1) b_k t^{k-2}$. Hence

$$\mathbf{P}_1 = tf'' + f' = - \sum k(k-1) b_k t^{k-1} - \sum k b_k t^{k-1} = - \sum k^2 b_k t^{k-1},$$

$$\mathbf{P}_3 = t^2 f'' - (p-1)tf' - f' = - \sum k(k-1) b_k t^k + (p-1) \sum k b_k t^k + \sum k b_k t^{k-1},$$

and $-(k-1)+(p-1) = p-k$ gives the stated form; likewise $\mathbf{Q} = t^2 f'' - (p-1)tf' = \sum k[(p-1)-(k-1)] b_k t^k = \sum k(p-k) b_k t^k$, which is ≥ 0 because $k < p$ for every $k \in E$ (this is also why $\mathbf{P}_3 \geq 0$). For $\mathbf{P}_2$,

$$\mathbf{P}_2 = p^2 t \left(a_0 - \sum b_k t^k \right) + (2p-1) \sum k b_k t^{k+1} - \sum k(k-1) b_k t^{k+1} - \sum k b_k t^{k-1},$$

and $-p^2 + (2p-1)k - k(k-1) = -(p^2 - 2pk + k^2) = -(p-k)^2$, which gives the stated form.

Continuity and (A5). $\varphi(r, s) = a_0 s^p - \sum b_k r^k s^{p-k}$ and every exponent $p - k > 0$, so each term tends to 0 as $s \downarrow 0$ with r fixed; thus φ extends continuously by $\varphi(r, 0) = 0$ and $\varphi(0, s) = a_0 s^p$. Homogeneity and smoothness on the open quadrant are clear (for $r > 0$, $r \mapsto r^k$ is C^∞ for any real k).

(A2). Each of $\varphi, \varphi_r, \varphi_s, \varphi_{rr}, \varphi_{rs}, \varphi_{ss}, \varphi_r/r, \varphi_s/s$ is a finite linear combination, with coefficients depending only on p, a_0 and $\max_k b_k$, of monomials $r^\alpha s^\beta$ with $\alpha \geq 0$ and $\alpha + \beta = p, p - 1, p - 2$ respectively. Explicitly:

$$\begin{aligned}\varphi_r &= -\sum k b_k r^{k-1} s^{p-k}, & \varphi_s &= p a_0 s^{p-1} - \sum (p-k) b_k r^k s^{p-k-1}, \\ \varphi_{rr} &= -\sum k(k-1) b_k r^{k-2} s^{p-k}, & \frac{\varphi_r}{r} &= -\sum k b_k r^{k-2} s^{p-k}, & \varphi_{rs} &= -\sum k(p-k) b_k r^{k-1} s^{p-k-1}, \\ \varphi_{ss} &= p(p-1) a_0 s^{p-2} - \sum (p-k)(p-k-1) b_k r^k s^{p-k-2}, & \frac{\varphi_s}{s} &= p a_0 s^{p-2} - \sum (p-k) b_k r^k s^{p-k-2}.\end{aligned}$$

All exponents α of r occurring above are ≥ 0 , because $k \geq 2$ for every $k \in E$. The exponents β of s occurring are, listed by order m :

- $m = 0$: $\beta = p$ and $\beta = p - k > 0$. Hence $\beta \geq 0$.
- $m = 1$: $\beta = p - k > 0$ (in φ_r), $\beta = p - 1 > 0$ and $\beta = p - k - 1$ (in φ_s). Since $\theta \geq k + 1 - p$ for every $k \in E$, we get $p - k - 1 \geq -\theta$; hence $\beta \geq -\theta$.
- $m = 2$: $\beta = p - k > 0$ (in $\varphi_{rr}, \varphi_r/r$), $\beta = p - k - 1 \geq -\theta$ (in φ_{rs}), $\beta = p - 2 \geq 0$ and $\beta = p - k - 2 \geq -1 - \theta$ (in $\varphi_{ss}, \varphi_s/s$). Hence $\beta \geq -(1 + \theta)$.

Note $\theta < 1$ because $k < p$ for all $k \in E$.

Now let $r^\alpha s^\beta$ with $\alpha \geq 0$, $\alpha + \beta = p - m$ and $\beta \geq -\gamma$ for some $\gamma \geq 0$. Since $r \leq \sigma$ and $s \leq \sigma$,

$$r^\alpha s^\beta \leq \sigma^\alpha s^\beta = \begin{cases} \sigma^\alpha s^\beta \leq \sigma^{p-m}, & \beta \geq 0, \\ \sigma^{p-m-\beta} s^\beta = \sigma^{p-m} \left(\frac{\sigma}{s}\right)^{-\beta} \leq \sigma^{p-m} \left(\frac{\sigma}{s}\right)^\gamma, & \beta < 0, \end{cases}$$

using $\sigma/s \geq 1$. Applying this with $(m, \gamma) = (0, 0)$, $(1, \theta)$ and $(2, 1 + \theta)$ gives exactly the three bounds of (A2), with M a constant depending only on p, a_0 and $\{b_k\}$. $\square$

Remark 2.9. Lemma 2.8 permits exponents k in the whole range $2 \leq k < p$; when $k > p - 1$ the derivative φ_s blows up like $s^{-(k+1-p)}$ as $s \downarrow 0$, which is precisely what the parameter θ in (A2) accommodates. This extra freedom is used at $p = \frac{16}{3}$ in Section 4.2 (there $E = \{2, 3, 4, 5\}$ and $5 > p - 1 = \frac{13}{3}$, so $\theta = \frac{2}{3}$); at $p = \frac{15}{2}$ in Section 4.3 all exponents satisfy $k \leq p - 1$ and $\theta = 0$.

Remark 2.10 (The quadratic coefficient). Assume $b_2 > 0$. As $t \downarrow 0$,

$$P_1 = -4b_2 t + o(t), \quad P_2 = (p^2 a_0 - 2b_2) t + o(t), \quad P_3 = 2b_2 t + o(t),$$

and hence

$$R = 4b_2(b_2 - p^2 a_0) t^2 + o(t^2).$$

Thus $R \geq 0$ near the origin forces $b_2 \geq p^2 a_0$. Both explicit profiles below saturate this necessary condition, $b_2 = p^2 a_0$, which makes the leading term vanish. This local observation is not, by itself, a global optimality statement.

Lemma 2.11 (majorization by weighted AM–GM). *In the setting of Lemma 2.8, let $t_0 > 0$, $\lambda > 0$ and $M > 0$ satisfy the two inequalities*

$$\lambda \left(a_0 - \sum_{k \in E} b_k \left(1 - \frac{k}{p} \right) t_0^k \right) \geq 1, \tag{M1}$$

$$M \geq \frac{\lambda}{p} \sum_{k \in E} k b_k t_0^{k-p}. \tag{M2}$$

Then $\lambda \varphi$ satisfies (A4) with $\Lambda := M^{1/p}$, i.e. $\lambda \varphi(r, s) \geq s^p - \Lambda^p r^p$ for all $r, s \geq 0$.

Proof. Step 1 (Young's inequality). For $0 < \vartheta \leq 1$ and $X \geq 0$,

$$X^\vartheta \leq \vartheta X + 1 - \vartheta.$$

Indeed $h(X) := \vartheta X + 1 - \vartheta - X^\vartheta$ satisfies $h(1) = 0$ and $h'(X) = \vartheta(1 - X^{\vartheta-1})$, which is ≤ 0 for $0 < X \leq 1$ and ≥ 0 for $X \geq 1$ (because $\vartheta - 1 \leq 0$); so h attains its minimum on $[0, \infty)$ at $X = 1$, where it vanishes. (At $X = 0$: $h(0) = 1 - \vartheta \geq 0$.)

Step 2. Let $k \in E$, so $0 < k < p$. Apply Step 1 with $\vartheta = k/p \in (0, 1)$ and $X = (t/t_0)^p$, noting $X^\vartheta = (t/t_0)^k$:

$$\left(\frac{t}{t_0}\right)^k \leq \frac{k}{p} \left(\frac{t}{t_0}\right)^p + 1 - \frac{k}{p} \implies t^k \leq \frac{k}{p} t_0^{k-p} t^p + \left(1 - \frac{k}{p}\right) t_0^k \quad (t \geq 0). \quad (2.8)$$

Step 3. Set $G(t) := \mathbf{M}t^p + \lambda f(t) - 1$. By (2.8) and $b_k \geq 0$,

$$\lambda \sum_k b_k t^k \leq \left(\frac{\lambda}{p} \sum_k k b_k t_0^{k-p}\right) t^p + \lambda \sum_k b_k \left(1 - \frac{k}{p}\right) t_0^k,$$

hence for all $t \geq 0$

$$G(t) = \mathbf{M}t^p + \lambda a_0 - \lambda \sum_k b_k t^k - 1 \geq \left[\mathbf{M} - \frac{\lambda}{p} \sum_k k b_k t_0^{k-p}\right] t^p + \left[\lambda \left(a_0 - \sum_k b_k \left(1 - \frac{k}{p}\right) t_0^k\right) - 1\right] \geq 0$$

by (M2) and (M1).

Step 4. For $s > 0$, dividing $\lambda\varphi(r, s) \geq s^p - \Lambda^p r^p$ by s^p and putting $t = r/s$ turns it into $\lambda f(t) \geq 1 - \mathbf{M}t^p$, i.e. $G(t) \geq 0$. For $s = 0$ it reads $0 = \lambda\varphi(r, 0) \geq -\Lambda^p r^p$, which is true. $\square$

Finally note that if φ satisfies (A1)–(A3) then so does $\lambda\varphi$ for any $\lambda > 0$ (all of (D0), (D1) are positively homogeneous in φ : $\mathcal{A}, \mathcal{B}, \mathcal{C}$ scale by λ , so $\mathcal{B}^2 \leq \mathcal{A}\mathcal{C}$ scales by λ^2 ; and (A2) holds with $\lambda\mathbf{M}$ and the same θ).

3. THE PROBABILISTIC MODEL AND THE BELLMAN PRINCIPLE

Fix $p \geq 2$, $f, g \in C_c^\infty(\mathbb{R}^2; \mathbb{C})$ and $T > 0$; write $U = e^{t\Delta} f$, $V = e^{t\Delta} g$ as in Section 0. Recall that U, V and all their space derivatives are bounded and continuous on $\mathbb{R}^2 \times [0, T]$, with $\|\partial^\alpha U(\cdot, t)\|_q \leq \|\partial^\alpha f\|_q$.

3.1. Brownian motion with generator Δ and the σ -finite measure ν . Let $\Omega = C([0, T]; \mathbb{R}^2)$ with the coordinate process $B_s(\omega) = \omega(s)$, the (augmented, right-continuous) filtration $(\mathcal{F}_s)_{0 \leq s \leq T}$, and for $x \in \mathbb{R}^2$ let P_x be the law of $x + \sqrt{2}W$, W a standard planar Brownian motion. Thus under P_x

$$d\langle B^i, B^j \rangle_s = 2\delta_{ij} ds, \quad P_x(B_s \in dy) = G_s(y - x) dy, \quad (3.1)$$

so the transition semigroup of B is exactly the heat semigroup $e^{s\Delta}$ and the generator is Δ . The map $x \mapsto E_x[\Phi]$ is Borel measurable for every bounded measurable Φ (standard for this Feller family).

Define the σ -finite measure on $(\Omega, \mathcal{F}_T)$

$$\nu(F) := \int_{\mathbb{R}^2} P_x(F) dx, \quad E_\nu[\Phi] := \int_{\mathbb{R}^2} E_x[\Phi] dx$$

(defined for $\Phi \geq 0$ measurable, and for $\Phi \in L^1(\nu)$). It is σ -finite because $\nu(\{B_0 \in B_R\}) = |B_R| < \infty$ and these sets exhaust Ω .

Lemma 3.1 (stationarity). *For every $s \in [0, T]$ and every Borel $\phi \geq 0$ on $\mathbb{R}^2$, $E_\nu[\phi(B_s)] = \int_{\mathbb{R}^2} \phi dx$.*

Proof. By (3.1) and Tonelli, $\int E_x[\phi(B_s)] dx = \iint \phi(y) G_s(y - x) dy dx = \int \phi(y) \left(\int G_s(y - x) dx\right) dy = \int \phi dy$, since G_s is a probability density. (For $s = 0$ this is trivial.) $\square$

3.2. The two martingales. Since $U \in C^{2,1}(\mathbb{R}^2 \times [0, T])$ and $\partial_t U = \Delta U$, Itô's formula applied under P_x to the real and imaginary parts of U gives, for $s \in [0, T]$,

$$X_s := U(B_s, T - s) = U(x, T) + \int_0^s \nabla U(B_r, T - r) \cdot dB_r, \quad (3.2)$$

because the drift is $(-\partial_t U + \Delta U)dr = 0$ (the Itô second-order term is $\frac{1}{2} \sum_{ij} \partial_i \partial_j U d\langle B^i, B^j \rangle = \Delta U dr$ by (3.1)). Similarly $N_s := V(B_s, T - s) = V(x, T) + \int_0^s \nabla V(B_r, T - r) \cdot dB_r$. Define the **transformed** martingale

$$Y_s := \int_0^s (A \nabla U)(B_r, T - r) \cdot dB_r, \quad Y_0 = 0. \quad (3.3)$$

All integrands are bounded continuous, so (3.2)–(3.3) are genuine $L^2(P_x)$ martingales for each x .

We record the complex Itô isometry: if $\mathbf{a}, \mathbf{b} : [0, T] \times \Omega \rightarrow \mathbb{C}^2$ are bounded adapted continuous, then by (3.1)

$$E_x \left[\left(\int_0^T \mathbf{a} \cdot dB \right) \overline{\left(\int_0^T \mathbf{b} \cdot dB \right)} \right] = 2 E_x \int_0^T \mathbf{a} \cdot \bar{\mathbf{b}} dr, \quad \mathbf{a} \cdot \bar{\mathbf{b}} = \sum_j \mathbf{a}_j \bar{\mathbf{b}}_j. \quad (3.4)$$

(Expand into real and imaginary parts and use the real Itô isometry $E[\int \mathbf{a} dB \int \mathbf{b} dB] = 2E \int \mathbf{a} \cdot \mathbf{b}$, valid by (3.1).)

Combining (3.4) with Lemma 3.1 and Tonelli, and using $|A\nabla U|^2 \leq 4|\nabla U|^2$ from (1.1) together with (1.4),

$$E_\nu |Y_T|^2 = 2 \int_0^T \int_{\mathbb{R}^2} |A\nabla U|^2 dx dt \leq 8 \int_0^\infty \int |\nabla U|^2 dx dt = 4\|f\|_2^2 < \infty, \quad (3.5)$$

likewise $E_\nu |N_T - N_0|^2 \leq \|g\|_2^2 < \infty$ and $E_\nu |N_0|^2 = \|V(\cdot, T)\|_2^2 \leq \|g\|_2^2 < \infty$. So $Y_T, N_0, N_T \in L^2(\nu)$.

The columns. Set, for $r \in [0, T]$,

$$\mathbf{a} := \partial_1 U(B_r, T - r), \quad \mathbf{b} := \partial_2 U(B_r, T - r), \quad \mathbf{c} := \mathbf{a} - i\mathbf{b}.$$

By Lemma 1.4, $A\nabla U = (\mathbf{c}, -i\mathbf{c})$. Hence, identifying $\mathbb{C} \simeq \mathbb{R}^2$, the $\mathbb{R}^4$ -valued continuous semimartingale

$$Z_s := (X_s, Y_s) \in \mathbb{R}^2 \times \mathbb{R}^2$$

satisfies $Z_s = Z_0 + \int_0^s \mathbb{M}_r dB_r$ with $Z_0 = (U(x, T), 0)$, where the j -th column of the real 4×2 matrix $\mathbb{M}_r$ is

$$v_1 = (\mathbf{a}, \mathbf{c}), \quad v_2 = (\mathbf{b}, -i\mathbf{c}), \quad \mathbf{c} = \mathbf{a} - i\mathbf{b}. \quad (3.6)$$

This is exactly the family appearing in Proposition 2.3 and Lemma 2.6(2).

3.3. The Bellman estimate.

Proposition 3.2. *Let $p \geq 2$ and let φ be (p, Λ) -admissible in the sense of Definition 2.4. Then for every $T > 0$ and every $f \in C_c^\infty$,*

$$\left(E_\nu |Y_T|^p \right)^{1/p} \leq \Lambda \|f\|_p. \quad (3.7)$$

Proof. Let $u(x, y) = \varphi(|x|, |y|)$, fix $\varepsilon > 0$ and let u_ε be as in Lemma 2.6. Choose $N > \|f\|_\infty$ and set $\tau_N := \inf\{s \in [0, T] : |Z_s| \geq N\} \wedge T$. Then continuity gives $|Z_{T \wedge \tau_N}| \leq N$. Itô's formula for $u_\varepsilon(Z_s)$ gives, P_x -a.s.,

$$u_\varepsilon(Z_{T \wedge \tau_N}) = u_\varepsilon(Z_0) + \underbrace{\int_0^{T \wedge \tau_N} \nabla u_\varepsilon(Z_r) \cdot dZ_r}_{=: \mathcal{M}} + \int_0^{T \wedge \tau_N} \sum_{j=1}^2 \langle D^2 u_\varepsilon(Z_r) v_j(r), v_j(r) \rangle dr,$$

where we used $d\langle Z^a, Z^b \rangle_r = 2 \sum_j \mathbb{M}_{aj} \mathbb{M}_{bj} dr$ (from (3.1)), so that the second-order term is $\frac{1}{2} \cdot 2 \sum_j \langle D^2 u_\varepsilon v_j, v_j \rangle dr$. By (3.6) and Lemma 2.6(2), that last integral is ≤ 0 . Hence, pointwise on Ω ,

$$u_\varepsilon(Z_{T \wedge \tau_N}) \leq u_\varepsilon(Z_0) + \mathcal{M}. \quad (3.8)$$

For $r < \tau_N$ we have $|Z_r| \leq N$, and ∇u_ε is continuous, hence bounded on the closed ball $\{|z| \leq N\}$ by some $C_{N, \varepsilon} < \infty$ (Lemma 2.6(1)); while $|\mathbb{M}_r| \leq C' |\nabla U(B_r, T - r)| \leq C' \|\nabla f\|_\infty$. Thus $\mathcal{M}$ is a stochastic integral of a bounded adapted integrand, so $\mathcal{M} \in L^2(P_x)$ and $E_x[\mathcal{M}] = 0$ for every x .

Fix $R > 0$, put $B_R := \{x : |x| < R\}$, and let μ_R denote the finite measure on $B_R \times \Omega$ induced by the probability kernel $x \mapsto P_x$:

$$\int_{B_R \times \Omega} F d\mu_R := \int_{B_R} E_x[F(x, \cdot)] dx \quad (F \geq 0).$$

Taking E_x in (3.8) (all terms are P_x -integrable, since $|u_\varepsilon(Z_{T \wedge \tau_N})| \leq C(N^p + \varepsilon^p)$) and then integrating over B_R gives

$$\int_{B_R} E_x[u_\varepsilon(Z_{T \wedge \tau_N})] dx \leq \int_{B_R} u_\varepsilon(U(x, T), 0) dx. \quad (3.9)$$

Let $\varepsilon \downarrow 0$ in (3.9). On the left, $u_\varepsilon \rightarrow u$ locally uniformly (Lemma 2.6(3)) and $|Z_{T \wedge \tau_N}| \leq N$, so the integrand converges pointwise and is bounded by $C(N^p + 1)$ on the finite measure space $(B_R \times \Omega, \mu_R)$; dominated convergence applies. On the right, $u_\varepsilon(U(x, T), 0) \rightarrow u(U(x, T), 0)$ pointwise with $|u_\varepsilon(U(x, T), 0)| \leq C(|U(x, T)|^p + 1) \in L^1(B_R)$. Hence

$$\int_{B_R} E_x[u(Z_{T \wedge \tau_N})] dx \leq \int_{B_R} u(U(x, T), 0) dx \leq 0,$$

the last inequality by (A5), since $u(x', 0) = \varphi(|x'|, 0) \leq 0$.

Now apply the majorization (A4), $u(x', y') \geq |y'|^p - \Lambda^p |x'|^p$:

$$\int_{B_R} E_x[|Y_{T \wedge \tau_N}|^p] dx \leq \Lambda^p \int_{B_R} E_x[|X_{T \wedge \tau_N}|^p] dx. \quad (3.10)$$

The process $X_s = U(B_s, T - s)$ is a continuous martingale bounded by $\|f\|_\infty$, so $|X_s|^p$ is a bounded submartingale (Jensen) and optional stopping gives $E_x[|X_{T \wedge \tau_N}|^p] \leq E_x[|X_T|^p] = E_x[|f(B_T)|^p]$. Therefore, by Lemma 3.1,

$$\int_{B_R} E_x[|X_{T \wedge \tau_N}|^p] dx \leq \int_{\mathbb{R}^2} E_x[|f(B_T)|^p] dx = \|f\|_p^p.$$

Letting $N \rightarrow \infty$ through values larger than $\|f\|_\infty$, continuity of each path on $[0, T]$ gives $\tau_N = T$ for all sufficiently large N , a.s.; so $|Y_{T \wedge \tau_N}| \rightarrow |Y_T|$ a.s. Fatou's lemma in (3.10) yields

$$\int_{B_R} E_x[|Y_T|^p] dx \leq \Lambda^p \|f\|_p^p,$$

and letting $R \uparrow \infty$ (monotone convergence) gives (3.7). $\square$

3.4. Proof of the Bellman principle.

Proposition 3.3. *Let $p \geq 2$ and let φ be (p, Λ) -admissible. Then for all $f, g \in C_c^\infty$ and every $T > 0$,*

$$\left| 2 \int_0^T \int_{\mathbb{R}^2} (A \nabla U) \cdot \nabla \bar{V} dx dt \right| \leq \Lambda \|f\|_p \|g\|_{p'}.$$

Proof. Using (3.4) under each P_x , then Lemma 3.1 and Fubini (legitimate since $\int_0^T \int |A \nabla U| |\nabla V| \leq 2 \int_0^T \int |\nabla U| |\nabla V| < \infty$ by Lemma 1.5 and Cauchy–Schwarz),

$$E_\nu[Y_T(\overline{N_T - N_0})] = 2 \int_{\mathbb{R}^2} E_x \int_0^T (A \nabla U) \cdot \nabla \bar{V}(B_r, T - r) dr dx = 2 \int_0^T \int_{\mathbb{R}^2} (A \nabla U) \cdot \nabla \bar{V} dx dt.$$

(The expectation on the left is absolutely convergent because Y_T and $N_T - N_0$ are in $L^2(\nu)$ by (3.5).) Moreover $E_x[Y_T \bar{N}_0] = \bar{V}(x, T) E_x[Y_T] = 0$ for each x , and $Y_T \bar{N}_0 \in L^1(\nu)$ by Cauchy–Schwarz in $L^2(\nu)$; hence $E_\nu[Y_T \bar{N}_0] = 0$ and

$$2 \int_0^T \int_{\mathbb{R}^2} (A \nabla U) \cdot \nabla \bar{V} dx dt = E_\nu[Y_T \bar{N}_T].$$

By Hölder's inequality on the measure space (Ω, ν) , Proposition 3.2, and Lemma 3.1 (which gives $\|N_T\|_{L^{p'}(\nu)} = (E_\nu[g(B_T)]^{p'})^{1/p'} = \|g\|_{p'}$),

$$|E_\nu[Y_T \bar{N}_T]| \leq \|Y_T\|_{L^p(\nu)} \|N_T\|_{L^{p'}(\nu)} \leq \Lambda \|f\|_p \|g\|_{p'}.$$

$\square$

Proof of the Bellman principle. Let φ be (p, Λ) -admissible, $p \geq 2$. Let $T \rightarrow \infty$ in Proposition 3.3: the integrand $(A \nabla U) \cdot \nabla \bar{V}$ is dominated in absolute value by $2|\nabla U| |\nabla V| \in L^1(\mathbb{R}^2 \times (0, \infty))$ (Lemma 1.5), so dominated convergence and (1.3) give

$$\left| \int (Bf) \bar{g} dx \right| \leq \Lambda \|f\|_p \|g\|_{p'} \quad (f, g \in C_c^\infty).$$

Then $Bf \in L^2 \subset L_{\text{loc}}^1$ and Lemma 1.2 (with $q = p'$, $h = Bf$, $M = \Lambda \|f\|_p$) gives $\|Bf\|_p \leq \Lambda \|f\|_p$. The estimate on C_c^∞ therefore extends uniquely to all of L^p , and $\|B\|_{L^p \rightarrow L^p} \leq \Lambda$. $\square$

4. THREE ADMISSIBLE BELLMAN FUNCTIONS

4.1. The rescaled Burkholder function. Fix $p \geq 2$ and write

$$q := p - 1 \geq 1, \quad \kappa := \sqrt{\frac{2p}{p-1}} = \sqrt{2 + \frac{2}{q}} \in (\sqrt{2}, 2], \quad \alpha_p := p \left(1 - \frac{1}{p}\right)^{p-1} = \frac{(p-1)^{p-1}}{p^{p-2}} > 0.$$

(Indeed $\kappa^2 = 2 + 2/q \leq 4$ because $q \geq 1$, and $\kappa^2 > 2$; so $1 < \sqrt{2} < \kappa \leq 2$.) Define

$$\Phi(\rho, s) := (s - q\rho)(\rho + s)^q \quad (\rho, s \geq 0), \quad \varphi^B(r, s) := \alpha_p \Phi(\kappa r, s). \quad (4.1)$$

Lemma 4.1. $\alpha_p \geq 1$ for every $p \geq 2$, with equality iff $p = 2$.

Proof. Put $X = p - 1 \geq 1$. Then $\alpha_p = (X + 1)\left(\frac{X}{X+1}\right)^X$, and $\alpha_p \geq 1$ is equivalent (take logarithms) to

$$\psi(X) := \ln(1 + X) - X \ln\left(1 + \frac{1}{X}\right) \geq 0.$$

We have $\psi(1) = \ln 2 - \ln 2 = 0$ and, using $\frac{d}{dX} \ln(1 + \frac{1}{X}) = \frac{1}{X+1} - \frac{1}{X} = -\frac{1}{X(X+1)}$,

$$\psi'(X) = \frac{1}{1+X} - \ln\left(1 + \frac{1}{X}\right) + X \cdot \frac{1}{X(X+1)} = \frac{2}{1+X} - \ln\left(1 + \frac{1}{X}\right).$$

Substituting $v = 1/X \in (0, 1]$, $\psi'(X) = \chi(v) := \frac{2v}{1+v} - \ln(1+v)$. Now $\chi(0) = 0$ and $\chi'(v) = \frac{2}{(1+v)^2} - \frac{1}{1+v} = \frac{1-v}{(1+v)^2} \geq 0$ for $v \in [0, 1]$, so $\chi \geq 0$ on $[0, 1]$, i.e. $\psi' \geq 0$ on $[1, \infty)$ (and $\psi'(X) > 0$ for $X > 1$). Hence $\psi(X) \geq \psi(1) = 0$ for $X \geq 1$. $\square$

Lemma 4.2 (majorization). For all $\rho, s \geq 0$: $\alpha_p \Phi(\rho, s) \geq s^p - (p-1)^p \rho^p$.

Proof. Both sides are positively homogeneous of degree p in (ρ, s) . The inequality is trivial at $(0, 0)$; otherwise divide by $(\rho + s)^p$ and set $t := \rho/(\rho + s) \in [0, 1]$, so $s/(\rho + s) = 1 - t$. The claim becomes $G(t) \geq 0$ on $[0, 1]$, where

$$G(t) := \alpha_p(1 - t - (p-1)t) - (1-t)^p + (p-1)^p t^p = \alpha_p(1 - pt) - (1-t)^p + (p-1)^p t^p.$$

Case $p = 2$. Then $\alpha_2 = 1$ and $G(t) = (1 - 2t) - (1 - t)^2 + t^2 = 0$.

Case $p > 2$. First,

$$G\left(\frac{1}{p}\right) = 0 - \left(1 - \frac{1}{p}\right)^p + \frac{(p-1)^p}{p^p} = -\left(\frac{p-1}{p}\right)^p + \left(\frac{p-1}{p}\right)^p = 0.$$

Next $G'(t) = -\alpha_p p + p(1-t)^{p-1} + p(p-1)^p t^{p-1}$, so, using $\alpha_p = (p-1)^{p-1}/p^{p-2}$,

$$G'\left(\frac{1}{p}\right) = -\alpha_p p + p\left(\frac{p-1}{p}\right)^{p-1} + p\frac{(p-1)^p}{p^{p-1}} = -\alpha_p p + \alpha_p + (p-1)\alpha_p = 0.$$

Finally $G''(t) = p(p-1)[(p-1)^p t^{p-2} - (1-t)^{p-2}]$, and for $t \in [0, 1]$, $p > 2$, taking $(p-2)$ -th roots of nonnegative numbers,

$$G''(t) \geq 0 \iff (p-1)^{p/(p-2)} t \geq 1-t \iff t \geq t_0 := \frac{1}{1 + (p-1)^{p/(p-2)}}.$$

Since $p-1 > 1$ and $\frac{p}{p-2} > 1$ we have $(p-1)^{p/(p-2)} > p-1$, hence $1 + (p-1)^{p/(p-2)} > p$ and therefore $t_0 < 1/p$. Thus G is concave on $[0, t_0]$ and convex on $[t_0, 1]$. On $[t_0, 1]$: G is convex with $G'(1/p) = 0$ and $1/p \in (t_0, 1)$, so $G \geq G(1/p) = 0$ there. On $[0, t_0]$: G is concave, hence $G \geq \min\{G(0), G(t_0)\}$, where $G(t_0) \geq 0$ by the previous sentence and $G(0) = \alpha_p - 1 \geq 0$ by Lemma 4.1. $\square$

Lemma 4.3 (derivatives). On $\{\rho > 0, s > 0\}$ put $\sigma := \rho + s > 0$, $\tau := \rho/\sigma \in (0, 1)$, and

$$K := p q \sigma^{p-3} > 0, \quad m := (q-1)\tau \geq 0, \quad J := \frac{1 - q\tau}{q(1-\tau)}.$$

Then

$$\Phi_{\rho\rho} = -K\sigma[1+m], \quad \Phi_{\rho s} = -K\sigma m, \quad \Phi_{ss} = K\sigma[1-m], \quad \frac{\Phi_\rho}{\rho} = -K\sigma, \quad \frac{\Phi_s}{s} = K\sigma J, \quad (4.2)$$

and $\Phi_\rho = -pq\rho\sigma^{p-2} \leq 0$. Moreover $J \leq \frac{1}{q}$.

Proof. Direct differentiation of $\Phi(\rho, s) = (s - q\rho)(\rho + s)^q$:

$$\Phi_\rho = -q\sigma^q + q(s - q\rho)\sigma^{q-1} = q\sigma^{q-1}[-\sigma + s - q\rho] = -pq\rho\sigma^{p-2},$$

$$\Phi_s = \sigma^q + q(s - q\rho)\sigma^{q-1} = \sigma^{p-2}[\sigma + qs - q^2\rho] = p\sigma^{p-2}[s - (p-2)\rho]$$

(using $1 - q^2 = 1 - (p-1)^2 = -p(p-2)$ and $\sigma = \rho + s$), and, differentiating once more,

$$\Phi_{\rho\rho} = -pq[\sigma^{p-2} + (p-2)\rho\sigma^{p-3}] = -K[q\rho + s], \quad \Phi_{\rho s} = -pq(p-2)\rho\sigma^{p-3} = -K(q-1)\rho,$$

$$\Phi_{ss} = p[(p-2)\sigma^{p-3}(s - (p-2)\rho) + \sigma^{p-2}] = p\sigma^{p-3}[(p-1)s - (p-1)(p-3)\rho] = K[s - (q-2)\rho]$$

(using $1 - (p-2)^2 = -(p-1)(p-3)$). Also $\frac{\Phi_\rho}{\rho} = -pq\sigma^{p-2} = -K\sigma$ and $\frac{\Phi_s}{s} = p\sigma^{p-2}\frac{s-(p-2)\rho}{s}$. Substituting $\rho = \tau\sigma$, $s = (1-\tau)\sigma$ and $p\sigma^{p-2} = K\sigma/q$: $\Phi_{\rho\rho} = -K\sigma[q\tau+1-\tau] = -K\sigma[1+m]$; $\Phi_{\rho s} = -K\sigma(q-1)\tau = -K\sigma m$; $\Phi_{ss} = K\sigma[(1-\tau) - (q-2)\tau] = K\sigma[1 - (q-1)\tau] = K\sigma[1-m]$; and

$$\frac{\Phi_s}{s} = p\sigma^{p-2}\frac{(1-\tau) - (q-1)\tau}{1-\tau} = \frac{K\sigma}{q} \cdot \frac{1-q\tau}{1-\tau} = K\sigma J.$$

Finally $J \leq 1/q \iff \frac{1-q\tau}{1-\tau} \leq 1 \iff 1-q\tau \leq 1-\tau \iff (q-1)\tau \geq 0$, which holds. $\square$

Proposition 4.4. *For every $p \geq 2$, φ^B is $(p, \sqrt{2p(p-1)})$ -admissible. Consequently, by the Bellman principle,*

$$\|B\|_{L^p \rightarrow L^p} \leq \sqrt{2p(p-1)} \quad (p \geq 2). \quad (4.3)$$

Proof. **(A1)** $\varphi^B(r, s) = \alpha_p(s - q\kappa r)(\kappa r + s)^q$ is continuous on $[0, \infty)^2$, positively homogeneous of degree p , and C^∞ where $\kappa r + s > 0$, in particular on the open quadrant.

(A5) $\varphi^B(r, 0) = \alpha_p(-q\kappa r)(\kappa r)^q \leq 0$.

(A4) By Lemma 4.2 with $\rho = \kappa r$, $\varphi^B(r, s) \geq s^p - (p-1)^p \kappa^p r^p = s^p - \Lambda^p r^p$ with $\Lambda = \kappa(p-1) = \sqrt{2p/(p-1)}(p-1) = \sqrt{2p(p-1)}$.

(A2) (with $\theta = 0$.) Since $\varphi^B(r, s) = \alpha_p \Phi(\rho, s)$ with $\rho = \kappa r$, we have

$$\varphi_{rr}^B = \alpha_p \kappa^2 \Phi_{\rho\rho}, \quad \varphi_{rs}^B = \alpha_p \kappa \Phi_{\rho s}, \quad \varphi_{ss}^B = \alpha_p \Phi_{ss}, \quad \frac{\varphi_r^B}{r} = \alpha_p \kappa^2 \frac{\Phi_\rho}{\rho}, \quad \frac{\varphi_s^B}{s} = \alpha_p \frac{\Phi_s}{s}. \quad (4.4)$$

Write $\sigma' = \kappa r + s \leq 2(r + s) = 2\sigma$. By (4.2) and $0 \leq m \leq q$, $\kappa \leq 2$: $|\varphi_{rr}^B|, |\varphi_{rs}^B|, |\varphi_{ss}^B|, |\varphi_r^B/r| \leq 4\alpha_p p q (1+q) \sigma'^{p-2}$, while

$$\left| \frac{\varphi_s^B}{s} \right| = \alpha_p p \sigma'^{p-2} \frac{|s - (p-2)\kappa r|}{s} \leq \alpha_p p (1 + 2(p-2)) \sigma'^{p-2} \left(1 + \frac{r}{s}\right).$$

Also $|\varphi^B| \leq \alpha_p q \sigma'^p$ (as $|s - q\kappa r| \leq q\sigma'$ for $q \geq 1$), $|\varphi_r^B| = \alpha_p p q \kappa^2 r \sigma'^{p-2} \leq 4\alpha_p p q \sigma'^{p-1}$ and $|\varphi_s^B| \leq \alpha_p p (1 + 2(p-2)) \sigma'^{p-1}$. Since $\sigma' \leq 2\sigma$ and $1 + r/s = \sigma/s$, (A2) holds with $\theta = 0$ and a suitable $M = M(p)$.

(A3) By (4.2) and (4.4), $\varphi_r^B = \alpha_p \kappa \Phi_\rho \leq 0$, which is (D0). Next, since $m \geq 0$, $\varphi_{rs}^B = -\alpha_p K \sigma' \kappa m \leq 0$, so $|\varphi_{rs}^B| = \alpha_p K \sigma' \kappa m$, and the entries of the drift matrix are ($\alpha_p K \sigma' > 0$ factored out):

$$\frac{\mathcal{A}}{\alpha_p K \sigma'} = -\kappa^2(1+m) - \kappa^2 = -\kappa^2(2+m), \quad \frac{\mathcal{B}}{\alpha_p K \sigma'} = \kappa m + \kappa^2 = \kappa(m + \kappa),$$

$$\frac{\mathcal{C}}{\alpha_p K \sigma'} = (1-m) + J - \kappa^2.$$

Clearly $\mathcal{A} < 0$. Also $\mathcal{C} \leq 0$, because by $J \leq 1/q$, $m \geq 0$ and $\kappa^2 = 2 + \frac{2}{q}$,

$$1 - m + J - \kappa^2 \leq 1 + \frac{1}{q} - 2 - \frac{2}{q} = -1 - \frac{1}{q} < 0.$$

It remains to check $\mathcal{B}^2 \leq \mathcal{A}\mathcal{C}$, i.e. $\kappa^2(m + \kappa)^2 \leq \kappa^2(2+m)(\kappa^2 - 1 + m - J)$, i.e.

$$D := (m+2)(\kappa^2 - 1 + m - J) - (m + \kappa)^2 \geq 0.$$

Expanding, $(m+2)(\kappa^2 - 1 + m) = m\kappa^2 + m^2 + m + 2\kappa^2 - 2$ and $(m + \kappa)^2 = m^2 + 2m\kappa + \kappa^2$, so

$$D = m\kappa^2 - 2m\kappa + m + \kappa^2 - 2 - (m+2)J = m(\kappa - 1)^2 + (\kappa^2 - 2) - (m+2)J.$$

Insert $\kappa^2 - 2 = \frac{2}{q}$, $m = (q-1)\tau$, $J = \frac{1-q\tau}{q(1-\tau)}$ and multiply by $q(1-\tau) > 0$:

$$q(1-\tau)D = q(1-\tau)(q-1)\tau(\kappa-1)^2 + 2(1-\tau) - ((q-1)\tau + 2)(1-q\tau).$$

The last two terms equal

$$2 - 2\tau - [(q-1)\tau - q(q-1)\tau^2 + 2 - 2q\tau] = \tau[-2 - (q-1) + 2q] + q(q-1)\tau^2 = (q-1)\tau[1 + q\tau].$$

Hence

$$q(1-\tau)D = (q-1)\tau \left[q(1-\tau)(\kappa-1)^2 + 1 + q\tau \right] \geq 0, \quad (4.5)$$

because $q \geq 1$, $\tau \in (0, 1)$ and $(\kappa-1)^2 \geq 0$. Thus $D \geq 0$, (D1) holds, and φ^B is admissible. $\square$

Remark 4.5 (criticality of κ). If κ is replaced by $\kappa' \in (1, \kappa)$ in (4.1) then, repeating the computation, at $\tau \rightarrow 0$ (i.e. $m \rightarrow 0$, $J \rightarrow 1/q$) one gets $D \rightarrow (\kappa'^2 - 2) - \frac{2}{q} < 0$. So κ cannot be lowered inside this one-parameter family; a genuinely different profile is needed, which is what Section 4.2–Section 4.3 provide.

4.2. A quintic profile at $p = \frac{16}{3}$. Let $p = \frac{16}{3}$, so $p - 1 = \frac{13}{3}$, $2p - 1 = \frac{29}{3}$, $p^2 = \frac{256}{9}$, and put

$$f(t) := 729 - 20736t^2 - 61440t^3 - 56320t^4 - 4320t^5, \quad (4.6)$$

$$\varphi(r, s) := s^{16/3} f(r/s) = 729 s^{16/3} - 20736 r^2 s^{10/3} - 61440 r^3 s^{7/3} - 56320 r^4 s^{4/3} - 4320 r^5 s^{1/3}.$$

This is Lemma 2.8 with $E = \{2, 3, 4, 5\} \subset [2, \frac{16}{3})$, $a_0 = 729$ and $(b_2, b_3, b_4, b_5) = (20736, 61440, 56320, 4320)$; note $b_2 = p^2 a_0 = \frac{256}{9} \cdot 729 = 256 \cdot 81 = 20736$, which saturates the necessary condition in Remark 2.10. The growth parameter is $\theta = \max\{0, 5 + 1 - \frac{16}{3}\} = \frac{2}{3} < 1$. Thus (A1), (A2), (A5) hold by Lemma 2.8.

Lemma 4.6 (drift). φ satisfies (D0) and (D1) at every (r, s) with $r, s > 0$; i.e. (A3) holds.

Proof. We use Lemma 2.8 with $p = \frac{16}{3}$. Since all $b_k \geq 0$ and all $k < p$, we already know $f' \leq 0$, $Q \geq 0$, $P_1 \leq 0$. Explicitly,

$$P_1 = -(4 \cdot 20736t + 9 \cdot 61440t^2 + 16 \cdot 56320t^3 + 25 \cdot 4320t^4) = -8t \Pi_1,$$

$$\Pi_1 := 13500t^3 + 112640t^2 + 69120t + 10368,$$

(check: $4 \cdot 20736 = 82944 = 8 \cdot 10368$, $9 \cdot 61440 = 552960 = 8 \cdot 69120$, $16 \cdot 56320 = 901120 = 8 \cdot 112640$, $25 \cdot 4320 = 108000 = 8 \cdot 13500$).

Next, with $p - k = \frac{10}{3}, \frac{7}{3}, \frac{4}{3}, \frac{1}{3}$ for $k = 2, 3, 4, 5$,

$$Q = \sum_k k(p-k)b_k t^k = 138240t^2 + 430080t^3 + \frac{901120}{3}t^4 + 7200t^5 = \frac{20}{3}t^2(1080t^3 + 45056t^2 + 64512t + 20736) \geq 0$$

(check: $\frac{20}{3} \cdot 20736 = 138240$, $\frac{20}{3} \cdot 64512 = 430080$, $\frac{20}{3} \cdot 45056 = \frac{901120}{3}$, $\frac{20}{3} \cdot 1080 = 7200$).

Also $\sum_k k b_k t^{k-1} = 41472t + 184320t^2 + 225280t^3 + 21600t^4$, so

$$P_3 = \sum_k k b_k t^{k-1} + Q = 41472t + 322560t^2 + 655360t^3 + \frac{965920}{3}t^4 + 7200t^5 = \frac{8}{3}t \Pi_3,$$

$$\Pi_3 := 2700t^4 + 120740t^3 + 245760t^2 + 120960t + 15552$$

(check: $\frac{8}{3} \cdot 15552 = 41472$, $\frac{8}{3} \cdot 120960 = 322560$, $\frac{8}{3} \cdot 245760 = 655360$, $\frac{8}{3} \cdot 120740 = \frac{965920}{3}$, $\frac{8}{3} \cdot 2700 = 7200$; and $184320 + 138240 = 322560$, $225280 + 430080 = 655360$, $21600 + \frac{901120}{3} = \frac{64800 + 901120}{3} = \frac{965920}{3}$).

For P_2 : $p^2 a_0 = \frac{256}{9} \cdot 729 = 20736$ and $(p - k)^2 b_k = \frac{100}{9} \cdot 20736 = 230400$, $\frac{49}{9} \cdot 61440 = \frac{1003520}{3}$, $\frac{16}{9} \cdot 56320 = \frac{901120}{9}$, $\frac{1}{9} \cdot 4320 = 480$ for $k = 2, 3, 4, 5$. Hence

$$\begin{aligned} P_2 &= 20736t - (41472t + 184320t^2 + 225280t^3 + 21600t^4) - (230400t^3 + \frac{1003520}{3}t^4 + \frac{901120}{9}t^5 + 480t^6) \\ &= -20736t - 184320t^2 - 455680t^3 - \frac{1068320}{3}t^4 - \frac{901120}{9}t^5 - 480t^6 = -\frac{8}{9}t \Pi_2 \leq 0, \end{aligned}$$

$$\Pi_2 := 540t^5 + 112640t^4 + 400620t^3 + 512640t^2 + 207360t + 23328$$

(check: $\frac{8}{9} \cdot 23328 = 20736$, $\frac{8}{9} \cdot 207360 = 184320$, $\frac{8}{9} \cdot 512640 = 455680$, $\frac{8}{9} \cdot 400620 = \frac{3204960}{9} = \frac{1068320}{3}$, $\frac{8}{9} \cdot 112640 = \frac{901120}{9}$, $\frac{8}{9} \cdot 540 = 480$; and $225280 + 230400 = 455680$, $21600 + \frac{1003520}{3} = \frac{64800 + 1003520}{3} = \frac{1068320}{3}$).

All the polynomials Π_1, Π_2, Π_3 have nonnegative coefficients, so indeed $P_1 \leq 0$, $P_2 \leq 0$ for $t > 0$.

Finally, since $P_1 P_2 = 8t \cdot \frac{8}{9}t \Pi_1 \Pi_2 = \frac{64t^2}{9} \Pi_1 \Pi_2$ and $P_3^2 = \frac{64t^2}{9} \Pi_3^2$,

$$R = P_1 P_2 - P_3^2 = \frac{64t^2}{9} (\Pi_1 \Pi_2 - \Pi_3^2).$$

Expanding the two degree-8 polynomials:

$$\begin{aligned} \Pi_1 \Pi_2 &= 7290000t^8 + 1581465600t^7 + 18133464400t^6 + 59837752320t^5 \\ &\quad + 89401835520t^4 + 63259263360t^3 + 22275440640t^2 + 3762339840t + 241864704, \\ \Pi_3^2 &= 7290000t^8 + 651996000t^7 + 15905251600t^6 + 59999308800t^5 \\ &\quad + 89691379200t^4 + 63209756160t^3 + 22275440640t^2 + 3762339840t + 241864704, \end{aligned}$$

so, subtracting coefficient by coefficient (the four lowest coefficients and the leading one cancel),

$$\Pi_1 \Pi_2 - \Pi_3^2 = 2t^3(464734800t^4 + 1114106400t^3 - 80778240t^2 - 144771840t + 24753600)$$

and therefore

$$R = \frac{128}{9} t^5 N(t), \quad N(t) := 464734800t^4 + 1114106400t^3 - 80778240t^2 - 144771840t + 24753600. \quad (4.7)$$

It remains to prove $N \geq 0$ on $[0, \infty)$. First we peel off, by an identity in t , a part all of whose coefficients are **nonnegative** (hence ≥ 0 for $t \geq 0$):

$$N(t) = 72000 M(t) + (46800t^4 + 50400t^3 + 5760t^2 + 20160t + 57600),$$

$$M(t) := 6454t^4 + 15473t^3 - 1122t^2 - 2011t + 343 \quad (4.8)$$

(check, coefficient by coefficient: $72000 \cdot 6454 = 464688000$ and $464688000 + 46800 = 464734800$; $72000 \cdot 15473 = 1114056000$ and $1114056000 + 50400 = 1114106400$; $72000 \cdot (-1122) = -80784000$ and $-80784000 + 5760 = -80778240$; $72000 \cdot (-2011) = -144792000$ and $-144792000 + 20160 = -144771840$; $72000 \cdot 343 = 24696000$ and $24696000 + 57600 = 24753600$).

Now M splits, again identically in t , into a part with nonnegative coefficients plus $q_1 + t q_2$ with two nonnegative quadratics:

$$M(t) = 6454t^4 + 44t^3 + q_1(t) + t q_2(t), \quad q_1(t) := 5680t^2 - 2782t + 343, \quad q_2(t) := 15429t^2 - 6802t + 771 \quad (4.9)$$

(check: the t^3 coefficient is $44 + 15429 = 15473$, the t^2 coefficient is $5680 - 6802 = -1122$, the t^1 coefficient is $-2782 + 771 = -2011$, and the constant term is 343). Both quadratics are nonnegative **on all of** $\mathbb{R}$, by their discriminants:

$$2782^2 = 7739524 \leq 4 \cdot 5680 \cdot 343 = 7792960, \quad 6802^2 = 46267204 \leq 4 \cdot 15429 \cdot 771 = 47583036.$$

Hence $q_1 \geq 0$ and $q_2 \geq 0$ everywhere, so by (4.9) $M(t) \geq 0$ for $t \geq 0$, so by the previous display $N(t) \geq 0$ for $t \geq 0$, and finally

$$R = \frac{128}{9} t^5 N(t) \geq 0 \quad (t > 0).$$

By Lemma 2.7, (D0) and (D1) hold. □

Lemma 4.7 (majorization). *Put*

$$t_0 := \left(\frac{47}{88}\right)^3 = \frac{103823}{681472}, \quad \lambda := \frac{19}{6185}, \quad M := 22830.$$

Then $\lambda\varphi$ satisfies (A4) with $\Lambda_1 := 22830^{3/16}$.

Proof. We verify (M1) and (M2) of Lemma 2.11. Write $y := \frac{47}{88}$, so $t_0 = y^3$, $t_0^k = y^{3k}$ and $t_0^{k-p} = y^{3k-16}$.

(M1). Here $1 - \frac{k}{p} = 1 - \frac{3k}{16}$, giving the weights $\frac{5}{8}, \frac{7}{16}, \frac{1}{4}, \frac{1}{16}$ for $k = 2, 3, 4, 5$, so

$$b_2 \left(1 - \frac{2}{p}\right) = 20736 \cdot \frac{5}{8} = 12960, \quad b_3 \cdot \frac{7}{16} = 61440 \cdot \frac{7}{16} = 26880, \quad b_4 \cdot \frac{1}{4} = 14080, \quad b_5 \cdot \frac{1}{16} = 270,$$

and

$$S := a_0 - \sum_k b_k \left(1 - \frac{k}{p}\right) t_0^k = 729 - 12960 y^6 - 26880 y^9 - 14080 y^{12} - 270 y^{15}.$$

Multiply by $88^{15} = 146973853900112596791703109632$ and use $y^{3j} = 47^{3j}/88^{3j}$:

$$88^{15} S = 729 \cdot 88^{15} - 12960 \cdot 47^6 \cdot 88^9 - 26880 \cdot 47^9 \cdot 88^6 - 14080 \cdot 47^{12} \cdot 88^3 - 270 \cdot 47^{15}.$$

The five integers are

$$\begin{aligned} &107143939493182083061151566921728, \quad 44211596576200490352388677304320, \\ &13970309212700369991666122096640, \quad 1114871892671277389595594588160, \\ &3257104054721499424453142610, \end{aligned}$$

whence $88^{15} S = 47843904707555223828076719789998$. Condition (M1), $\lambda S \geq 1$ with $\lambda = \frac{19}{6185}$, is therefore equivalent to

$$19 \cdot 47843904707555223828076719789998 \geq 6185 \cdot 146973853900112596791703109632,$$

i.e. to

$$909034189443549252733457676009962 \geq 909033286372196411156683733073920,$$

which is true. (The ratio is $1.00000099 \dots$)

(M2). Here $\frac{\lambda}{p} = \frac{19}{6185} \cdot \frac{3}{16} = \frac{57}{98960}$ and

$$\Sigma := \sum_k k b_k t_0^{k-p} = 41472 y^{-10} + 184320 y^{-7} + 225280 y^{-4} + 21600 y^{-1},$$

using $2b_2 = 41472$, $3b_3 = 184320$, $4b_4 = 225280$, $5b_5 = 21600$. Since $y^{-1} = \frac{88}{47}$, multiplying by $47^{10} = 52599132235830049$ gives

$$\begin{aligned} 47^{10}\Sigma &= 41472 \cdot 88^{10} + 184320 \cdot 47^3 \cdot 88^7 \\ &\quad + 225280 \cdot 47^6 \cdot 88^4 + 21600 \cdot 47^9 \cdot 88. \end{aligned}$$

whose four summands are

$$\begin{aligned} &1154999247706192481353728, \quad 782068404177362611077120, \\ &145626520759631683256320, \quad 2127243203273739513600. \end{aligned}$$

Their sum is

$$47^{10}\Sigma = 2084821415846460515200768.$$

Condition (M2), $M \geq \frac{57}{98960}\Sigma$, is therefore equivalent to

$$22830 \cdot 98960 \cdot 47^{10} \geq 57 \cdot 2084821415846460515200768.$$

The two sides are, respectively,

$$\begin{aligned} &22830 \cdot 98960 \cdot 47^{10} = 118834947177898241847583200, \\ &57 \cdot 2084821415846460515200768 = 118834820703248249366443776. \end{aligned}$$

Thus (M2) holds. Numerically, $\frac{\lambda}{p}\Sigma = 22829.9757 \dots \leq 22830$.

By Lemma 2.11, $\lambda\varphi$ satisfies (A4) with $\Lambda_1 = M^{1/p} = 22830^{3/16}$. □

Proposition 4.8.

$$\|B\|_{L^{16/3} \rightarrow L^{16/3}} \leq \Lambda_1 = 22830^{3/16}. \quad (4.10)$$

Proof. φ satisfies (A1), (A2) (with $\theta = \frac{2}{3}$) and (A5) by Lemma 2.8, and (A3) by Lemma 4.6. Multiplying by $\lambda = \frac{19}{6185} > 0$ preserves (A1), (A2), (A3), (A5), and Lemma 4.7 gives (A4) with Λ_1 . So $\lambda\varphi$ is $(\frac{16}{3}, \Lambda_1)$ -admissible, and the Bellman principle applies. □

Numerically $\Lambda_1 = 6.5647752 \dots$, whereas $\sqrt{2p(p-1)} = \sqrt{2 \cdot \frac{16}{3} \cdot \frac{13}{3}} = \sqrt{\frac{416}{9}} = 6.7986 \dots$; so (4.10) improves (4.3) at $p = \frac{16}{3}$.

4.3. A sextic profile at $p = \frac{15}{2}$. Let $p = \frac{15}{2}$, so $p-1 = \frac{13}{2}$, $2p-1 = 14$, $p^2 = \frac{225}{4}$, and put

$$f(t) := 12 - 675t^2 - 3300t^3 - 6600t^4 - 5423t^5 - 2640t^6, \quad (4.11)$$

$$\varphi(r, s) := s^{15/2} f(r/s) = 12s^{15/2} - 675r^2s^{11/2} - 3300r^3s^{9/2} - 6600r^4s^{7/2} - 5423r^5s^{5/2} - 2640r^6s^{3/2}.$$

This is Lemma 2.8 with $E = \{2, 3, 4, 5, 6\} \subset [2, \frac{15}{2})$, $a_0 = 12$ and $(b_2, \dots, b_6) = (675, 3300, 6600, 5423, 2640)$; again $b_2 = p^2 a_0 = \frac{225}{4} \cdot 12 = 675$. All exponents satisfy $k \leq 6 \leq p-1 = \frac{13}{2}$, so here $\theta = 0$. Thus (A1), (A2), (A5) hold by Lemma 2.8.

Lemma 4.9 (drift). φ satisfies (D0) and (D1) at every (r, s) with $r, s > 0$.

Proof. By Lemma 2.8 with $p = \frac{15}{2}$:

$$P_1 = -(4 \cdot 675t + 9 \cdot 3300t^2 + 16 \cdot 6600t^3 + 25 \cdot 5423t^4 + 36 \cdot 2640t^5) = -5t \Theta_1,$$

$$\Theta_1 := 19008t^4 + 27115t^3 + 21120t^2 + 5940t + 540$$

(check: $5 \cdot 540 = 2700 = 4 \cdot 675$; $5 \cdot 5940 = 29700 = 9 \cdot 3300$; $5 \cdot 21120 = 105600 = 16 \cdot 6600$; $5 \cdot 27115 = 135575 = 25 \cdot 5423$; $5 \cdot 19008 = 95040 = 36 \cdot 2640$).

With $p-k = \frac{11}{2}, \frac{9}{2}, \frac{7}{2}, \frac{5}{2}, \frac{3}{2}$ for $k = 2, \dots, 6$,

$$\begin{aligned} Q &= \sum_k k(p-k)b_k t^k \\ &= 7425t^2 + 44550t^3 + 92400t^4 + \frac{135575}{2}t^5 + 23760t^6 \\ &= \frac{55}{2}t^2(864t^4 + 2465t^3 + 3360t^2 + 1620t + 270) \geq 0. \end{aligned}$$

(check: $\frac{55}{2} \cdot 270 = 7425$, $\frac{55}{2} \cdot 1620 = 44550$, $\frac{55}{2} \cdot 3360 = 92400$, $\frac{55}{2} \cdot 2465 = \frac{135575}{2}$, $\frac{55}{2} \cdot 864 = 23760$).

Also $\sum_k k b_k t^{k-1} = 1350t + 9900t^2 + 26400t^3 + 27115t^4 + 15840t^5$, hence

$$P_3 = \sum_k k b_k t^{k-1} + Q = 1350t + 17325t^2 + 70950t^3 + 119515t^4 + \frac{167255}{2}t^5 + 23760t^6 = \frac{5}{2}t \Theta_3,$$

$$\Theta_3 := 9504t^5 + 33451t^4 + 47806t^3 + 28380t^2 + 6930t + 540$$

(check: $\frac{5}{2} \cdot 540 = 1350$, $\frac{5}{2} \cdot 6930 = 17325$, $\frac{5}{2} \cdot 28380 = 70950$, $\frac{5}{2} \cdot 47806 = 119515$, $\frac{5}{2} \cdot 33451 = \frac{167255}{2}$, $\frac{5}{2} \cdot 9504 = 23760$; and $9900 + 7425 = 17325$, $26400 + 44550 = 70950$, $27115 + 92400 = 119515$, $15840 + \frac{135575}{2} = \frac{31680+135575}{2} = \frac{167255}{2}$).

For P_2 : $p^2 a_0 = 675$, and $(p-k)^2 b_k = \frac{121}{4} \cdot 675 = \frac{81675}{4}$, $\frac{81}{4} \cdot 3300 = 66825$, $\frac{49}{4} \cdot 6600 = 80850$, $\frac{25}{4} \cdot 5423 = \frac{135575}{4}$, $\frac{9}{4} \cdot 2640 = 5940$ for $k = 2, \dots, 6$. Hence

$$\begin{aligned} P_2 &= 675t - (1350t + 9900t^2 + 26400t^3 + 27115t^4 + 15840t^5) \\ &\quad - \left(\frac{81675}{4}t^3 + 66825t^4 + 80850t^5 + \frac{135575}{4}t^6 + 5940t^7 \right) \\ &= -675t - 9900t^2 - \frac{187275}{4}t^3 - 93940t^4 - 96690t^5 - \frac{135575}{4}t^6 - 5940t^7 = -\frac{5}{4}t \Theta_2 \leq 0, \end{aligned}$$

$$\Theta_2 := 4752t^6 + 27115t^5 + 77352t^4 + 75152t^3 + 37455t^2 + 7920t + 540$$

(check: $\frac{5}{4} \cdot 540 = 675$, $\frac{5}{4} \cdot 7920 = 9900$, $\frac{5}{4} \cdot 37455 = \frac{187275}{4}$, $\frac{5}{4} \cdot 75152 = 93940$, $\frac{5}{4} \cdot 77352 = 96690$, $\frac{5}{4} \cdot 27115 = \frac{135575}{4}$, $\frac{5}{4} \cdot 4752 = 5940$; and $26400 + \frac{81675}{4} = \frac{105600+81675}{4} = \frac{187275}{4}$, $27115 + 66825 = 93940$, $15840 + 80850 = 96690$).

All of $\Theta_1, \Theta_2, \Theta_3$ have nonnegative coefficients, so $P_1 \leq 0$ and $P_2 \leq 0$ for $t > 0$.

Since $P_1 P_2 = 5t \cdot \frac{5}{4}t \Theta_1 \Theta_2 = \frac{25t^2}{4} \Theta_1 \Theta_2$ and $P_3^2 = \frac{25t^2}{4} \Theta_3^2$,

$$R = \frac{25t^2}{4} (\Theta_1 \Theta_2 - \Theta_3^2).$$

Expanding the two degree-10 polynomials:

$$\begin{aligned} \Theta_1 \Theta_2 &= 90326016t^{10} + 644252400t^9 + 2305892281t^8 + 4126784376t^7 + 4546994540t^6 \\ &\quad + 3227458905t^5 + 1504237680t^4 + 444977280t^3 + 78675300t^2 + 7484400t + 291600, \\ \Theta_3^2 &= 90326016t^{10} + 635836608t^9 + 2027665849t^8 + 3737764052t^7 + 4315817836t^6 \\ &\quad + 3187363740t^5 + 1504142640t^4 + 444977280t^3 + 78675300t^2 + 7484400t + 291600, \end{aligned}$$

so

$$\Theta_1 \Theta_2 - \Theta_3^2 = t^4 (8415792t^5 + 278226432t^4 + 389020324t^3 + 231176704t^2 + 40095165t + 95040)$$

and therefore

$$R = \frac{25}{4} t^6 (8415792t^5 + 278226432t^4 + 389020324t^3 + 231176704t^2 + 40095165t + 95040) \geq 0 \quad (4.12)$$

for $t > 0$. Lemma 2.7 gives (D0), (D1). □

Lemma 4.10 (majorization). *Put*

$$t_0 := \left(\frac{8}{25} \right)^2 = \frac{64}{625}, \quad \lambda := \frac{105}{454}, \quad M := 22865000.$$

Then $\lambda \varphi$ satisfies (A_4) with $\Lambda_2 := 22865000^{2/15}$.

Proof. Write $w := \frac{8}{25}$, so $t_0 = w^2$, $t_0^k = w^{2k}$, $t_0^{k-p} = w^{2k-15}$.

(M1). Here $1 - \frac{k}{p} = 1 - \frac{2k}{15}$, giving weights $\frac{11}{15}, \frac{3}{5}, \frac{7}{15}, \frac{1}{3}, \frac{1}{5}$ for $k = 2, \dots, 6$, so

$$675 \cdot \frac{11}{15} = 495, \quad 3300 \cdot \frac{3}{5} = 1980, \quad 6600 \cdot \frac{7}{15} = 3080, \quad 5423 \cdot \frac{1}{3} = \frac{5423}{3}, \quad 2640 \cdot \frac{1}{5} = 528,$$

$$S = 12 - 495w^4 - 1980w^6 - 3080w^8 - \frac{5423}{3}w^{10} - 528w^{12}.$$

Multiply by $3 \cdot 25^{12} = 178813934326171875$:

$$3 \cdot 25^{12} S = 36 \cdot 25^{12} - 1485 \cdot 8^4 \cdot 25^8 - 5940 \cdot 8^6 \cdot 25^6 - 9240 \cdot 8^8 \cdot 25^4 - 5423 \cdot 8^{10} \cdot 25^2 - 1584 \cdot 8^{12},$$

whose six terms are

$$2145767211914062500, \quad 928125000000000000, \quad 380160000000000000,$$

6055526400000000, 3639313694720000, 108851651149824,
giving $3 \cdot 25^{12}S = 773178782568192676$. Condition (M1), $\frac{105}{454}S \geq 1$, is therefore equivalent to

$$105 \cdot 773178782568192676 \geq 454 \cdot 178813934326171875,$$

i.e. to

$$81183772169660230980 \geq 81181526184082031250,$$

which is true. (The ratio is $1.0000276\dots$)

(M2). Here $\frac{\lambda}{p} = \frac{105}{454} \cdot \frac{2}{15} = \frac{7}{227}$ and, using $2b_2 = 1350$, $3b_3 = 9900$, $4b_4 = 26400$, $5b_5 = 27115$, $6b_6 = 15840$,

$$\Sigma = \sum_k k b_k t_0^{k-p} = 1350 w^{-11} + 9900 w^{-9} + 26400 w^{-7} + 27115 w^{-5} + 15840 w^{-3}.$$

Since $w^{-1} = \frac{25}{8}$, multiplying by $8^{11} = 8589934592$ gives

$$8^{11}\Sigma = 1350 \cdot 25^{11} + 9900 \cdot 8^2 \cdot 25^9 + 26400 \cdot 8^4 \cdot 25^7 + 27115 \cdot 8^6 \cdot 25^5 + 15840 \cdot 8^8 \cdot 25^3,$$

whose five terms are

$$3218650817871093750, \quad 2416992187500000000, \quad 660000000000000000,$$

$$694144000000000000, \quad 41523609600000000,$$

totalling $8^{11}\Sigma = 6369209766331093750$. Condition (M2), $M \geq \frac{7}{227}\Sigma$, is therefore equivalent to

$$22865000 \cdot 227 \cdot 8^{11} \geq 7 \cdot 6369209766331093750.$$

Now $227 \cdot 8^{11} = 1949915152384$ and $22865000 \cdot 1949915152384 = 44584809959260160000$, while $7 \cdot 6369209766331093750 = 44584468364317656250$; since

$$44584809959260160000 \geq 44584468364317656250,$$

(M2) holds. (Indeed $\frac{\lambda}{p}\Sigma = 22864824.8\dots \leq 22865000$.)

By Lemma 2.11, $\lambda\varphi$ satisfies (A4) with $\Lambda_2 = M^{1/p} = 22865000^{2/15}$. □

Proposition 4.11.

$$\|B\|_{L^{15/2} \rightarrow L^{15/2}} \leq \Lambda_2 = 22865000^{2/15}. \quad (4.13)$$

Proof. Identical to Proposition 4.8: φ satisfies (A1), (A2) (with $\theta = 0$) and (A5) by Lemma 2.8, and (A3) by Lemma 4.9; multiplying by $\lambda = \frac{105}{454} > 0$ preserves these, and Lemma 4.10 supplies (A4). Apply the Bellman principle. □

Numerically $\Lambda_2 = 9.5768531\dots$, whereas $\sqrt{2 \cdot \frac{15}{2} \cdot \frac{13}{2}} = \sqrt{\frac{195}{2}} = 9.8742\dots$

4.4. Endpoint summary. Collecting (0.2), Lemma 1.3, (4.3), (4.10) and (4.13):

Endpoint theorem. $\|B\|_p = \|B\|_{p'}$ for all $1 < p < \infty$; $\|B\|_2 = 1$; and

$$\|B\|_p \leq \sqrt{2p(p-1)} \quad (p \geq 2), \quad \|B\|_{16/3} \leq 22830^{3/16}, \quad \|B\|_{15/2} \leq 22865000^{2/15}.$$

5. INTERPOLATION AND THE UNIFORM CONSTANT

Throughout this section $u := 1.523958$. We must show

$$\|B\|_p \leq u(p-1) \quad \text{for all } p \geq 2; \quad (5.1)$$

the range $1 < p < 2$ then follows from $\|B\|_p = \|B\|_{p'}$ (Lemma 1.3) and $p^* - 1 = p' - 1$.

5.1. A single interpolation interval.

Lemma 5.1. *Let $2 \leq p_0 < p_1 < \infty$ and let $L_0 < L_1$ be real numbers with $\|B\|_{p_0} \leq e^{L_0}$ and $\|B\|_{p_1} \leq e^{L_1}$. Put*

$$\hat{c} := \frac{L_1 - L_0}{2\left(\frac{1}{p_0} - \frac{1}{p_1}\right)} > 0, \quad K := L_0 - 2\hat{c}\left(\frac{1}{2} - \frac{1}{p_0}\right).$$

Then for every $p \in [p_0, p_1]$, writing $t := 1 - \frac{2}{p} \in [1 - \frac{2}{p_0}, 1 - \frac{2}{p_1}]$,

$$\ln \frac{\|B\|_p}{p-1} \leq K + \hat{c}t - 2 \operatorname{artanh} t. \quad (5.2)$$

Proof. B is a linear operator bounded on L^{p_0} and on L^{p_1} (Section 0), so Riesz–Thorin applies: if $\frac{1}{p} = \frac{1-\vartheta}{p_0} + \frac{\vartheta}{p_1}$ with $\vartheta \in [0, 1]$, then $\|B\|_p \leq \|B\|_{p_0}^{1-\vartheta} \|B\|_{p_1}^{\vartheta} \leq e^{(1-\vartheta)L_0 + \vartheta L_1}$. Writing $x := 1/p$, the exponent is the value at x of the affine function through $(1/p_0, L_0)$ and $(1/p_1, L_1)$, namely

$$L(x) = L_0 + m\left(x - \frac{1}{p_0}\right), \quad m := \frac{L_1 - L_0}{\frac{1}{p_1} - \frac{1}{p_0}} = -2\hat{c} < 0.$$

With $t = 1 - 2x$, i.e. $x = \frac{1-t}{2}$,

$$L(x) = L_0 + m\left(\frac{1}{2} - \frac{1}{p_0}\right) - \frac{m}{2}t = K + \hat{c}t.$$

Finally $p - 1 = \frac{1+t}{1-t}$, so $\ln(p - 1) = \ln \frac{1+t}{1-t} = 2 \operatorname{artanh} t$, and $\ln \|B\|_p - \ln(p - 1) \leq K + \hat{c}t - 2 \operatorname{artanh} t$. $\square$

Lemma 5.2 (Legendre transform). *Let $\hat{c} > 2$ and $S(t) := \hat{c}t - 2 \operatorname{artanh} t$ on $[0, 1)$. Put $g := \sqrt{\hat{c}(\hat{c} - 2)} > 0$ and $t^* := g/\hat{c} = \sqrt{1 - 2/\hat{c}} \in (0, 1)$. Then S is strictly increasing on $[0, t^*]$, strictly decreasing on $[t^*, 1)$, and*

$$\max_{[0, 1)} S = S(t^*) = g - 2 \operatorname{artanh}(t^*).$$

Proof. $S'(t) = \hat{c} - \frac{2}{1-t^2}$ is strictly decreasing on $[0, 1)$, with $S'(0) = \hat{c} - 2 > 0$ and $S'(t) \rightarrow -\infty$ as $t \uparrow 1$. Its unique zero is at $1 - t^2 = 2/\hat{c}$, i.e. $t = t^*$; note $\hat{c}t^* = \hat{c}\sqrt{1 - 2/\hat{c}} = \sqrt{\hat{c}^2 - 2\hat{c}} = g$, so $S(t^*) = g - 2 \operatorname{artanh}(t^*)$. $\square$

Lemma 5.3 (shift of artanh , and the logarithm identity). *1. For $\alpha, \beta \in (-1, 1)$, $\operatorname{artanh} \alpha - \operatorname{artanh} \beta = \operatorname{artanh} \frac{\alpha - \beta}{1 - \alpha\beta}$.*

2. For all $A, B > 0$, $\ln A = \ln B + 2 \operatorname{artanh} \frac{A - B}{A + B}$.

3. $\operatorname{artanh} \frac{3}{5} = \ln 2$ and $\operatorname{artanh} \frac{2}{3} = \frac{1}{2} \ln 5$.

Proof. (1) Both sides are $\frac{1}{2} \ln$ of $\frac{(1+\alpha)(1-\beta)}{(1-\alpha)(1+\beta)}$, since $\frac{1+\frac{\alpha-\beta}{1-\alpha\beta}}{1-\frac{\alpha-\beta}{1-\alpha\beta}} = \frac{1-\alpha\beta+\alpha-\beta}{1-\alpha\beta-\alpha+\beta} = \frac{(1+\alpha)(1-\beta)}{(1-\alpha)(1+\beta)}$. (2) Put $x := \frac{A-B}{A+B} \in (-1, 1)$. Then $\frac{1+x}{1-x} = \frac{(A+B)+(A-B)}{(A+B)-(A-B)} = \frac{2A}{2B} = \frac{A}{B}$, so $2 \operatorname{artanh} x = \ln \frac{A}{B}$. (3) $\operatorname{artanh} \frac{3}{5} = \frac{1}{2} \ln \frac{1+3/5}{1-3/5} = \frac{1}{2} \ln 4 = \ln 2$ and $\operatorname{artanh} \frac{2}{3} = \frac{1}{2} \ln \frac{1+2/3}{1-2/3} = \frac{1}{2} \ln 5$. $\square$

Combining Lemmas 5.1–5.3: if $\hat{c} > 2$ and t^* lies in the t -interval of the range, then (5.1) holds on $[p_0, p_1]$ as soon as

$$K + g \leq \ln u + 2 \operatorname{artanh}(\beta) + 2 \operatorname{artanh}(\delta), \quad \delta := \frac{t^* - \beta}{1 - \beta t^*}, \quad (5.3)$$

for any convenient base point $\beta \in (0, 1)$. Indeed, by Lemma 5.2 and Lemma 5.3(1), for every t in the range

$$\ln \frac{\|B\|_p}{p-1} \leq K + g - 2 \operatorname{artanh} t^* = K + g - 2 \operatorname{artanh} \beta - 2 \operatorname{artanh} \delta,$$

and (5.3) says exactly that the right-hand side is $\leq \ln u$. We shall use $\beta = \frac{3}{5}$ (so $2 \operatorname{artanh} \beta = 2 \ln 2$ and the right side of (5.3) begins with $\ln(4u)$) and $\beta = \frac{2}{3}$ (so $2 \operatorname{artanh} \beta = \ln 5$ and it begins with $\ln(5u)$).

5.2. The three ranges and their data. We use Theorem A at the two base points $p = \frac{16}{3}$ and $p = \frac{15}{2}$, together with $\|B\|_2 = 1$:

$$L_{(0)} := 0, \quad L_{(1)} := \frac{3}{16} \ln 22830, \quad L_{(2)} := \frac{2}{15} \ln 22865000. \quad (5.4)$$

These are legitimate by (0.2), (4.10) and (4.13): $\|B\|_2 = 1 = e^{L_{(0)}}$, $\|B\|_{16/3} \leq 22830^{3/16} = e^{L_{(1)}}$, $\|B\|_{15/2} \leq 22865000^{2/15} = e^{L_{(2)}}$.

Reduction of the two logarithms to $\ln 2, \ln 3, \ln 5, \ln 7$. By Lemma 5.3(2):

- $2^9 \cdot 3^7 = 512 \cdot 2187 = 1119744$, and $22830 \cdot 49 = 1118670$, so with $A = 22830$, $B = \frac{1119744}{49}$ we get $A + B = \frac{1118670 + 1119744}{49} = \frac{2238414}{49}$, $A - B = \frac{1118670 - 1119744}{49} = -\frac{1074}{49}$, and $\frac{1074}{2238414} = \frac{179}{373069}$ (divide by 6), whence

$$\ln 22830 = 9 \ln 2 + 7 \ln 3 - 2 \ln 7 - 2 \operatorname{artanh} \frac{179}{373069}. \quad (5.5)$$

- $2^5 \cdot 3^5 \cdot 5^5 = 32 \cdot 243 \cdot 3125 = 24300000$, and with $A = 22865000$, $B = 24300000$: $A - B = -1435000$, $A + B = 47165000$, and $\frac{1435000}{47165000} = \frac{287}{9433}$ (divide by 5000), whence

$$\ln 22865000 = 5 \ln 2 + 5 \ln 3 + 5 \ln 5 - 2 \operatorname{artanh} \frac{287}{9433}. \quad (5.6)$$

- $u = 1.523958$, $B = \frac{3}{2}$: $A - B = 0.023958$, $A + B = 3.023958$, so $\frac{A-B}{A+B} = \frac{23958}{3023958} = \frac{3993}{503993}$ (divide by 6) and

$$\begin{aligned} \ln u &= \ln 3 - \ln 2 + 2 \operatorname{artanh} \frac{3993}{503993}, \\ \ln(4u) &= \ln 2 + \ln 3 + 2 \operatorname{artanh} \frac{3993}{503993}, \quad \ln(5u) = \ln 3 - \ln 2 + \ln 5 + 2 \operatorname{artanh} \frac{3993}{503993}. \end{aligned} \quad (5.7)$$

The three ranges.

range	p -interval	t -interval	$\hat{c}$	K	base β
I	$[2, \frac{16}{3}]$	$[0, \frac{5}{8}]$	$\frac{8}{5} L_{(1)} = \frac{3}{16} \ln 22830$	0	$\frac{3}{5}$
II	$[\frac{16}{3}, \frac{15}{2}]$	$[\frac{5}{8}, \frac{11}{15}]$	$\frac{120}{13} (L_{(2)} - L_{(1)})$	$L_{(1)} - \frac{5}{8} \hat{c}$	$\frac{2}{3}$
III	$[\frac{15}{2}, \infty)$	—	—	—	—

(These are Lemma 5.1. Range I: $\frac{1}{p_0} - \frac{1}{p_1} = \frac{1}{2} - \frac{3}{16} = \frac{5}{16}$, so $\hat{c} = \frac{L_{(1)}}{2 \cdot \frac{5}{16}} = \frac{8}{5} L_{(1)} = \frac{8}{5} \cdot \frac{3}{16} \ln 22830 = \frac{3}{10} \ln 22830$, and $K = 0 - 2\hat{c}(\frac{1}{2} - \frac{1}{2}) = 0$; the t -interval is $[1 - \frac{2}{2}, 1 - \frac{2 \cdot 3}{16}] = [0, \frac{5}{8}]$. Range II: $\frac{3}{16} - \frac{2}{15} = \frac{45-32}{240} = \frac{13}{240}$, so $\hat{c} = \frac{L_{(2)} - L_{(1)}}{13/120} = \frac{120}{13} (L_{(2)} - L_{(1)})$, and $K = L_{(1)} - 2\hat{c}(\frac{1}{2} - \frac{3}{16}) = L_{(1)} - \frac{5}{8} \hat{c}$; the t -interval is $[\frac{5}{8}, 1 - \frac{4}{15}] = [\frac{5}{8}, \frac{11}{15}]$.)

5.3. Range III: $p \geq \frac{15}{2}$. By (4.3), $\frac{\|B\|_p}{p-1} \leq \sqrt{\frac{2p}{p-1}}$, and $\frac{2p}{p-1} = 2 + \frac{2}{p-1}$ is decreasing in p , so for $p \geq \frac{15}{2}$ (where $p-1 \geq \frac{13}{2}$)

$$\left(\frac{\|B\|_p}{p-1} \right)^2 \leq 2 + \frac{2}{13/2} = \frac{30}{13}.$$

Since $13u^2 = 13 \cdot (1.523958)^2 = 13 \cdot 2.322447985764 = 30.191823814932 > 30$, we get $\|B\|_p \leq u(p-1)$ on $[\frac{15}{2}, \infty)$.

5.4. Rigorous numerical bounds. For $0 < x < 1$ the series $\operatorname{artanh} x = \sum_{k \geq 0} \frac{x^{2k+1}}{2k+1}$ has positive terms, so every partial sum is a lower bound, and

$$\sum_{k=0}^{n-1} \frac{x^{2k+1}}{2k+1} \leq \operatorname{artanh} x \leq \sum_{k=0}^{n-1} \frac{x^{2k+1}}{2k+1} + \frac{x^{2n+1}}{(2n+1)(1-x^2)} \quad (5.8)$$

(bounding the tail $\sum_{k \geq n} \frac{x^{2k+1}}{2k+1} \leq \frac{1}{2n+1} \sum_{k \geq n} x^{2k+1} = \frac{x^{2n+1}}{(2n+1)(1-x^2)}$ by a geometric series). Using

$$\ln 2 = 2 \operatorname{artanh} \frac{1}{3}, \quad \ln 3 = \ln 2 + 2 \operatorname{artanh} \frac{1}{5}, \quad \ln 5 = 2 \ln 2 + 2 \operatorname{artanh} \frac{1}{9}, \quad \ln 7 = 3 \ln 2 - 2 \operatorname{artanh} \frac{1}{15}$$

(each identity is Lemma 5.3(2): $\frac{1+x}{1-x} = 2, \frac{3}{2}, \frac{5}{4}, \frac{8}{7}$ for $x = \frac{1}{3}, \frac{1}{5}, \frac{1}{9}, \frac{1}{15}$, so $\ln 2 = 2 \operatorname{artanh} \frac{1}{3}$, $\ln 3 = \ln 2 + \ln \frac{3}{2}$, $\ln 5 = \ln 4 + \ln \frac{5}{4}$, $\ln 7 = \ln 8 - \ln \frac{8}{7}$), together with (5.8) applied with $n = 40, 25, 20, 18$ terms respectively, one obtains

$$\begin{aligned} \ln 2 &\in (0.6931471804, 0.6931471806), & \ln 3 &\in (1.0986122884, 1.0986122888), \\ \ln 5 &\in (1.6094379120, 1.6094379126), & \ln 7 &\in (1.9459101484, 1.9459101492). \end{aligned} \quad (5.9)$$

Also, directly from (5.8) with $n = 2, 4, 3$ terms respectively,

$$\operatorname{artanh} \frac{179}{373069} \in (0.0004798040, 0.0004798041),$$

$$\operatorname{artanh} \frac{287}{9433} \in (0.0304344966, 0.0304344967), \quad \operatorname{artanh} \frac{3993}{503993} \in (0.0079228948, 0.0079228949). \quad (5.10)$$

(For instance $x = \frac{179}{373069} = 0.00047980400\dots$, so $x + \frac{x^3}{3} = 0.00047980404\dots$ and the $n = 2$ tail $\frac{x^5}{5(1-x^2)}$ is $< 10^{-17}$; and for $x = \frac{3993}{503993} = 0.0079227290\dots$, $x + \frac{x^3}{3} + \frac{x^5}{5} = 0.0079228948607\dots$ with $n = 3$ tail $< 10^{-15}$.)

All numbers below are obtained from (5.9)–(5.10) by **interval arithmetic on ten decimals**: each intermediate quantity is enclosed in an interval whose endpoints are ten-decimal rationals, obtained by rounding **outwards** the exact rational bounds produced by $+$, $-$, $\times$, $\div$, and, for square roots, by the elementary rule “ $a \leq \sqrt{X} \leq b$ whenever $a^2 \leq X$ and $b^2 \geq X$ ” with ten-decimal rationals a, b . Every single step is a finite rational computation, and every displayed interval below can be re-verified by these four operations alone.

From (5.5), (5.6), (5.4), (5.7) and (5.9)–(5.10):

$$\ln 22830 \in (10.0358307358, 10.0358307422), \quad \ln 22865000 \in (16.9451179106, 16.9451179168), \quad (5.11)$$

$$L_{(1)} \in (1.8817182629, 1.8817182642), \quad L_{(2)} \in (2.2593490547, 2.2593490556), \quad (5.12)$$

$$\ln u \in (0.4213108974, 0.4213108982),$$

$$\ln(4u) \in (1.8076052584, 1.8076052592), \quad (5.13)$$

$$\ln(5u) \in (2.0307488094, 2.0307488108).$$

(For example, the lower end of (5.11) is $9 \cdot 0.6931471804 + 7 \cdot 1.0986122884 - 2 \cdot 1.9459101492 - 2 \cdot 0.0004798041 = 6.2383246236 + 7.6902860188 - 3.8918202984 - 0.0009596082 = 10.0358307358$, and the upper end is $9 \cdot 0.6931471806 + 7 \cdot 1.0986122888 - 2 \cdot 1.9459101484 - 2 \cdot 0.0004798040 = 10.0358307422$; then $L_{(1)} = \frac{3}{16} \ln 22830$ gives (5.12) after outward rounding — e.g. $\frac{3}{16} \cdot 10.0358307358 = 1.8817182629625$, which rounds down to 1.8817182629. Likewise $\ln u = \ln 3 - \ln 2 + 2 \operatorname{artanh} \frac{3993}{503993}$ has lower end $1.0986122884 - 0.6931471806 + 2 \cdot 0.0079228948 = 0.4213108974$.)

Finally we record the bound for the shift term used twice below: if $|\delta| \leq x$ with $0 < x \leq \frac{1}{10}$, then by (5.8) with $n = 1$ and oddness of artanh ,

$$|\operatorname{artanh} \delta| \leq \operatorname{artanh} x \leq x + \frac{x^3}{3(1-x^2)}. \quad (5.14)$$

5.5. Range I: $2 \leq p \leq \frac{16}{3}$. Here $K = 0$ and $\hat{c} = \frac{3}{10} \ln 22830$, so by (5.11)

$$\hat{c} \in (3.0107492207, 3.0107492227).$$

In particular $\hat{c} > 2$. Since $X \mapsto X(X-2)$ is increasing for $X > 1$,

$$g^2 = \hat{c}(\hat{c}-2) \in (3.0107492207 \cdot 1.0107492207, 3.0107492227 \cdot 1.0107492227) \subset (3.0431124285, 3.0431124366),$$

and, since $1.7444518991^2 \leq 3.0431124285$ while $1.7444519015^2 \geq 3.0431124366$,

$$g \in (1.7444518991, 1.7444519015). \quad (5.15)$$

Hence $t^* = g/\hat{c} \in (\frac{1.7444518991}{3.0107492227}, \frac{1.7444519015}{3.0107492207}) \subset (0.5794079048, 0.5794079061)$, which lies in $(0, \frac{5}{8})$; so Lemma 5.2 gives the maximum of S on the t -interval $[0, \frac{5}{8}]$ at t^* . With $\beta = \frac{3}{5}$, the map $t \mapsto \frac{t-3/5}{1-3t/5}$ is increasing, so

$$\delta \in (-0.0315657689, -0.0315657667), \quad |\delta| \leq x := 0.0315657689.$$

By (5.14): $x^2 \leq 0.0009963978$, $x^3 \leq 0.0000314521$, and $\frac{x^3}{3(1-x^2)} \leq \frac{0.0000314521}{3 \cdot 0.9990036022} \leq 0.0000104945$, so $\operatorname{artanh} x \leq 0.0315657689 + 0.0000104945 \leq 0.0315762634$ and

$$2 \operatorname{artanh} \delta \geq -0.0631525268.$$

Therefore, by (5.13),

$$\ln(4u) + 2 \operatorname{artanh} \delta \geq 1.8076052584 - 0.0631525268 = 1.7444527316,$$

while (recall $K = 0$)

$$K + g = g \leq 1.7444519015 < 1.7444527316.$$

This is (5.3), so $\|B\|_p \leq u(p-1)$ for $2 \leq p \leq \frac{16}{3}$. The margin is $8.3 \cdot 10^{-7}$.

5.6. Range II: $\frac{16}{3} \leq p \leq \frac{15}{2}$. By (5.12),

$$L_{(2)} - L_{(1)} \in (2.2593490547 - 1.8817182642, 2.2593490556 - 1.8817182629) \subset (0.3776307905, 0.3776307927),$$

$$\hat{c} = \frac{120}{13}(L_{(2)} - L_{(1)}) \in (3.4858226815, 3.4858227019),$$

$$\begin{aligned} K &= L_{(1)} - \frac{5}{8}\hat{c} \\ &\in (1.8817182629 - 0.625 \cdot 3.4858227019, \\ &\quad 1.8817182642 - 0.625 \cdot 3.4858226815) \\ &\subset (-0.2969209258, -0.2969209117). \end{aligned}$$

Again $\hat{c} > 2$, and

$$g^2 = \hat{c}(\hat{c} - 2) \in (3.4858226815 \cdot 1.4858226815, 3.4858227019 \cdot 1.4858227019) \subset (5.1793144038, 5.1793145053),$$

so, since $2.2758107135^2 \leq 5.1793144038$ and $2.2758107359^2 \geq 5.1793145053$,

$$g \in (2.2758107135, 2.2758107359).$$

Hence $t^* = g/\hat{c} \in (\frac{2.2758107135}{3.4858227019}, \frac{2.2758107359}{3.4858226815}) \subset (0.6528762097, 0.6528762201)$, which lies in $(\frac{5}{8}, \frac{11}{15}) = (0.625, 0.7333\dots)$; so Lemma 5.2 applies on the t -interval $[\frac{5}{8}, \frac{11}{15}]$. With $\beta = \frac{2}{3}$, the increasing map $t \mapsto \frac{t-2/3}{1-2t/3}$ gives

$$\delta \in (-0.0244187284, -0.0244187095), \quad |\delta| \leq x := 0.0244187284.$$

By (5.14): $x^2 \leq 0.0005962743$, $x^3 \leq 0.0000145603$, and $\frac{x^3}{3(1-x^2)} \leq \frac{0.0000145603}{3 \cdot 0.9994037257} \leq 0.0000048564$, so $\operatorname{artanh} x \leq 0.0244187284 + 0.0000048564 = 0.0244235848$ and

$$2 \operatorname{artanh} \delta \geq -0.0488471696.$$

Therefore, by (5.13),

$$\ln(5u) + 2 \operatorname{artanh} \delta \geq 2.0307488094 - 0.0488471696 = 1.9819016398,$$

while

$$K + g \leq -0.2969209117 + 2.2758107359 = 1.9788898242 < 1.9819016398.$$

This is (5.3) with $\beta = \frac{2}{3}$ (whose $2 \operatorname{artanh} \beta = \ln 5$, so $\ln u + 2 \operatorname{artanh} \beta = \ln(5u)$), so $\|B\|_p \leq u(p-1)$ for $\frac{16}{3} \leq p \leq \frac{15}{2}$. The margin is $3.01 \cdot 10^{-3}$.

5.7. Completion of the proof. Combining the three ranges proves

$$\|B\|_{L^p \rightarrow L^p} \leq 1.523958(p-1), \quad p \geq 2.$$

For $1 < p < 2$, transposition gives $\|B\|_p = \|B\|_{p'}$, while $p^* - 1 = p' - 1$. This completes the proof of the main theorem.

6. CONCLUDING REMARKS

- (1) The binding interpolation interval is $[2, 16/3]$. High-precision evaluation of the explicit interpolation envelopes gives

$$\sup_{2 \leq p \leq 16/3} \frac{\|B\|_p}{p-1} \leq 1.5239567163458625\dots,$$

with the maximum at $p = 4.7552011223\dots$. The other two envelopes are $1.5193749633\dots$ and $\sqrt{30/13} = 1.5191090506\dots$. These decimal values are included only for orientation; the proof of the theorem uses the outward-rounded rational interval certificate in Section 5.

- (2) The condition $b_2 \geq p^2 a_0$ in Remark 2.10 is merely a necessary local condition for the polynomial drift determinant near $t = 0$. The two profiles used here choose equality, but no claim of global optimality within all polynomial profiles, or within all Bellman functions satisfying the drift criterion, is made.
- (3) The parameter $\theta < 1$ in the growth axiom is essential at $p = 16/3$: the monomial $r^5 s^{1/3}$ produces a second derivative of order $s^{-5/3}$, which remains locally integrable in the two-dimensional y variable. Compactly supported convolution then preserves the drift inequality by a direct distributional argument.

- (4) The proof identifies two possible sources of further improvement: a stronger pointwise use of the joint geometry of the radial and tangential components, or new endpoint Bellman profiles outside the present generalized-polynomial family. The current argument establishes only the explicit coefficient stated in the main theorem and does not assert sharpness of the method.

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